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Windows Log Monitoring & Threat Detection with Splunk

Project Summary

This project is a hands-on **Security Operations Center (SOC) lab** designed to demonstrate how a SOC analyst can monitor Windows security events, collect logs, identify suspicious activities, investigate security incidents, and create detections using **Splunk SIEM**.

The lab uses a **Windows virtual machine** as the monitored endpoint. Security activities are intentionally generated in the lab environment, such as failed authentication attempts, creation of a new user account, addition of a user to the local Administrators group, and PowerShell/process execution.

These Windows events are collected and forwarded to **Splunk Enterprise** using the **Splunk Universal Forwarder**. The collected logs are then analyzed using **Splunk Search Processing Language (SPL)** to identify potentially suspicious behavior.

The project follows a basic SOC workflow:

Windows Endpoint



Windows Security Logs



Splunk Universal Forwarder



Splunk Enterprise (SIEM)



SPL Detection Rules



SOC Investigation



SOC Dashboard

Understand these 6 concepts:

1.Log

A record of something that happened.

Example:

User: student

Action: Login failed

Time: 10:32 PM

Source: 192.168.x.x

2. Event

A particular activity recorded by Windows.

3. SIEM

A system that collects and analyzes security logs.

Wazuh will be your SIEM/security monitoring platform.

4. Alert

A suspicious event identified by a detection rule.

Example:

5 failed logins in a short period

↓

Suspicious

↓

Alert

5.Incident investigation

You investigate:

Who?

What?

When?

Where?

How?

Is it actually malicious?

What should we do?

6. MITRE ATT&CK

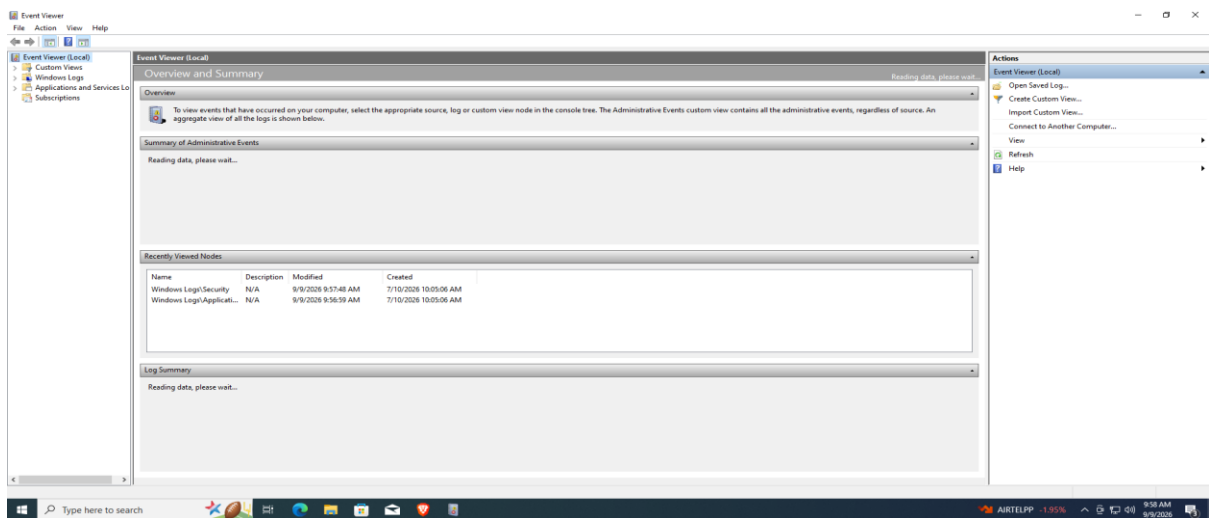
A framework for describing attacker techniques.

You will use it to explain what a detected behaviour represents.

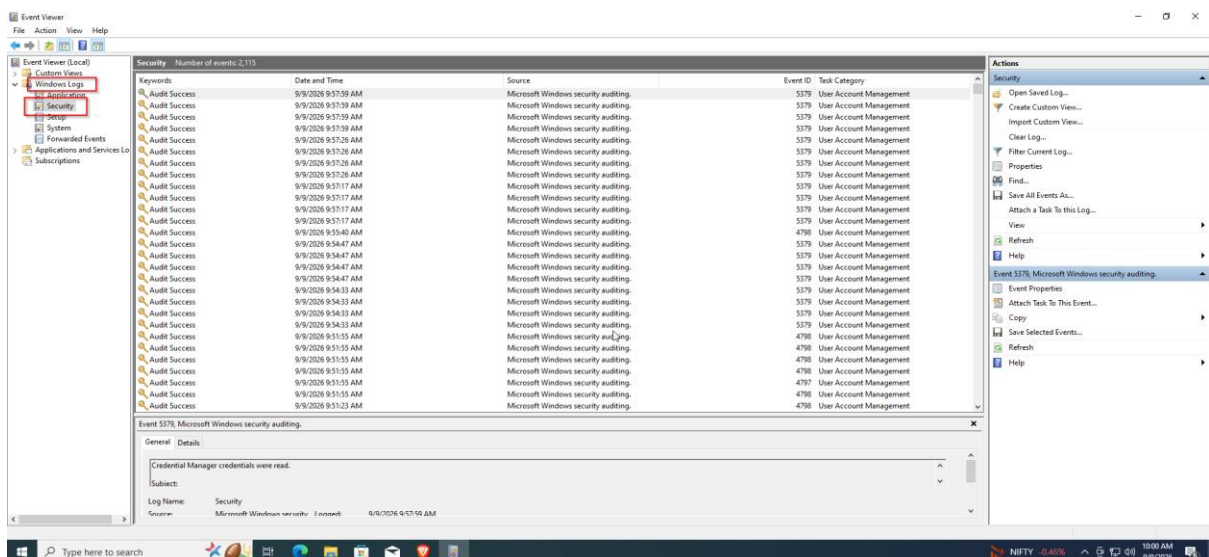
Phase 1 - Install VirtualBox and Create Windows VM

Phase 2 - Learn Windows Event Viewer (SOC Basic Investigation)

Step-1: Open Event Viewer.



Step-2: Here, we can view and monitor all Windows Security events.

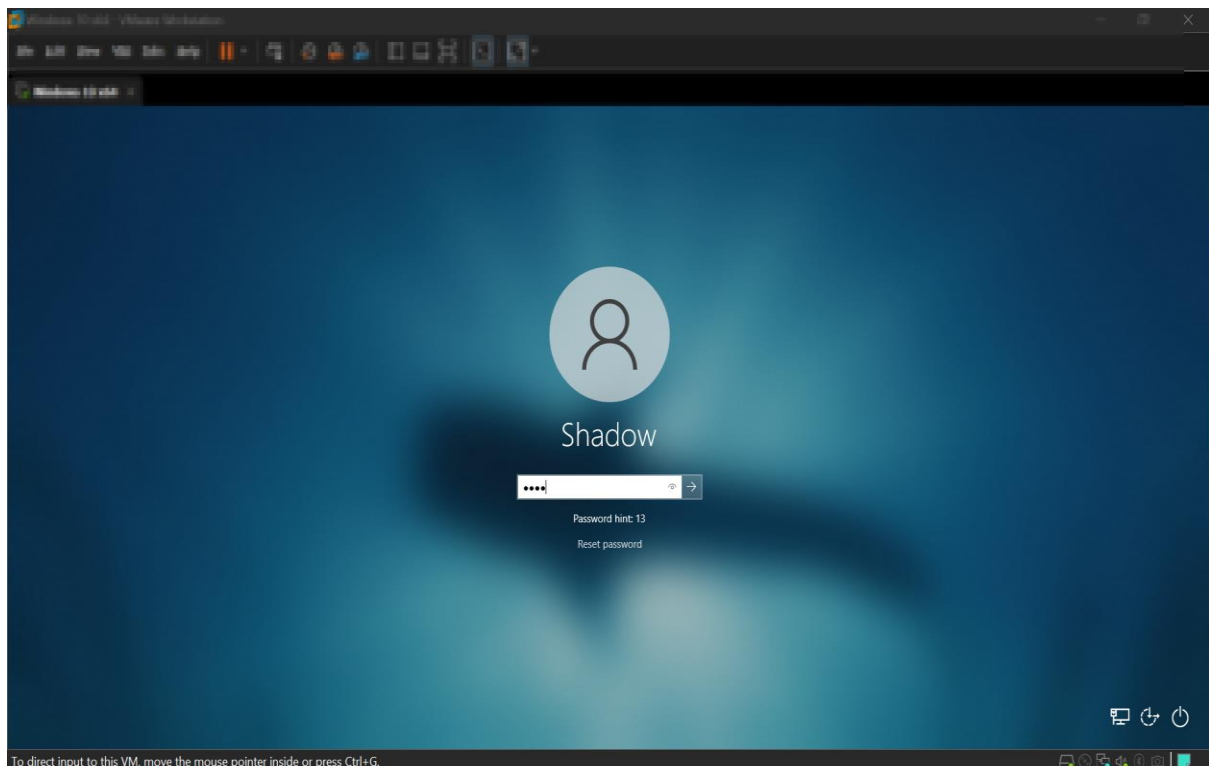


Step-3: Event ID

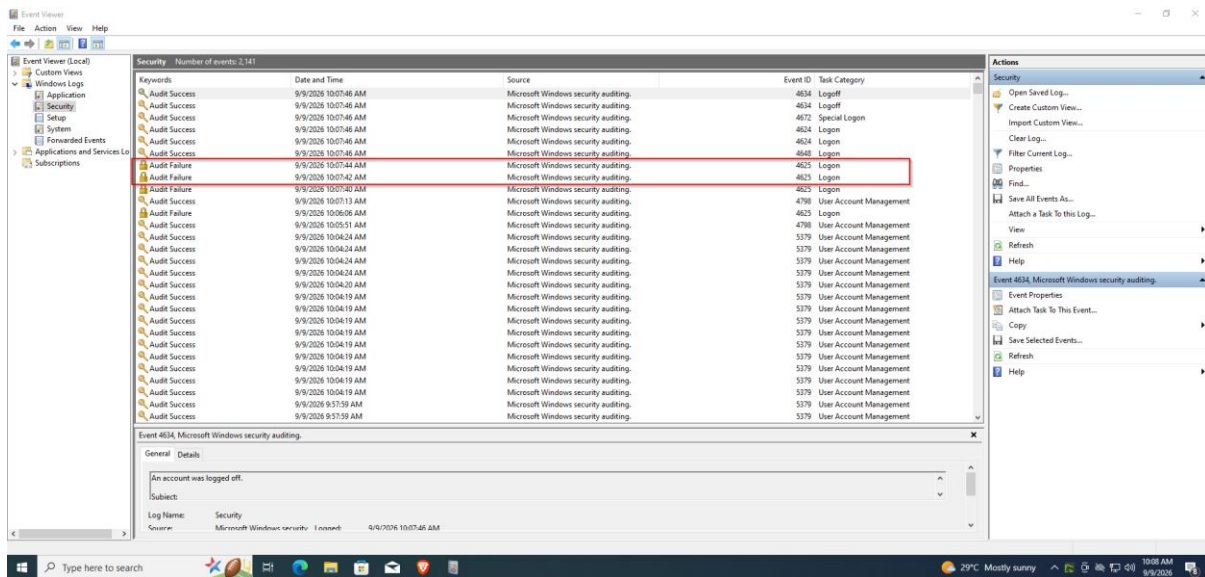
Event ID	Meaning
4624	Successful logon
4625	Failed logon
4720	User account created
4722	User account enabled
4726	User account deleted
4732	Member added to local security-enabled group
4688	New process created

Phase 3 - Generate first security events

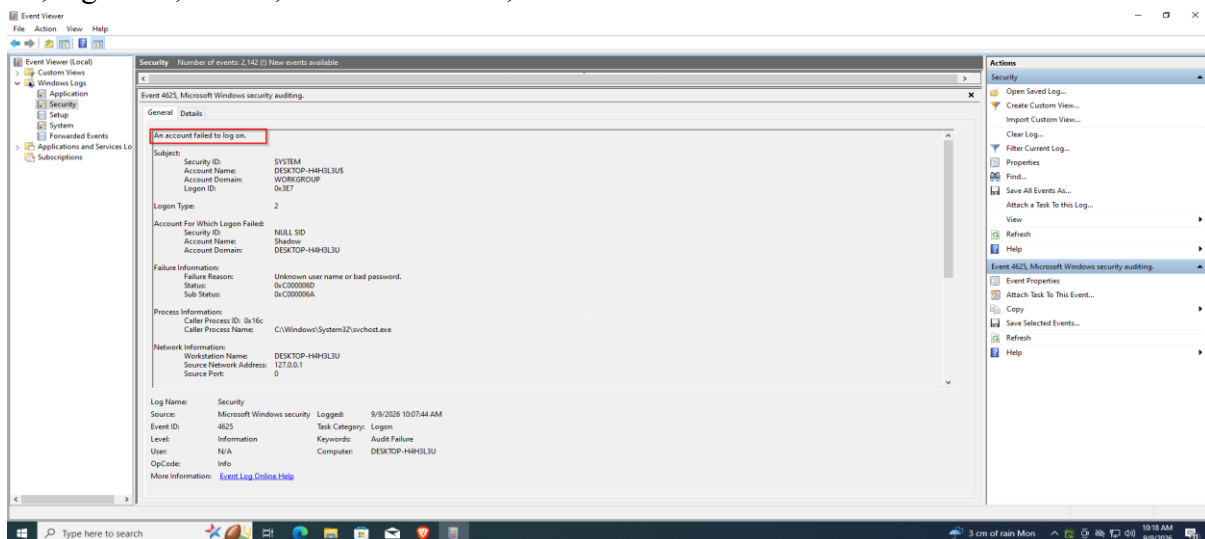
Step-4: Here, we can enter an incorrect password to generate a failed login attempt.



Step-5: Here, you can see that the logon attempt was audited as a failure because an incorrect password was entered. The event ID is 4625, and the audit category is Logon.



Step-6: Here, you can see all the details of the event in the General tab, including the event ID, log name, source, time of the event, and audit status.



First investigation:

Time: 10:07:44

Username: Shadow

Computer: DESKTOP-H4H3L3U

Logon Type: 2

Source IP: 127.0.0.1

Failure reason: %2313

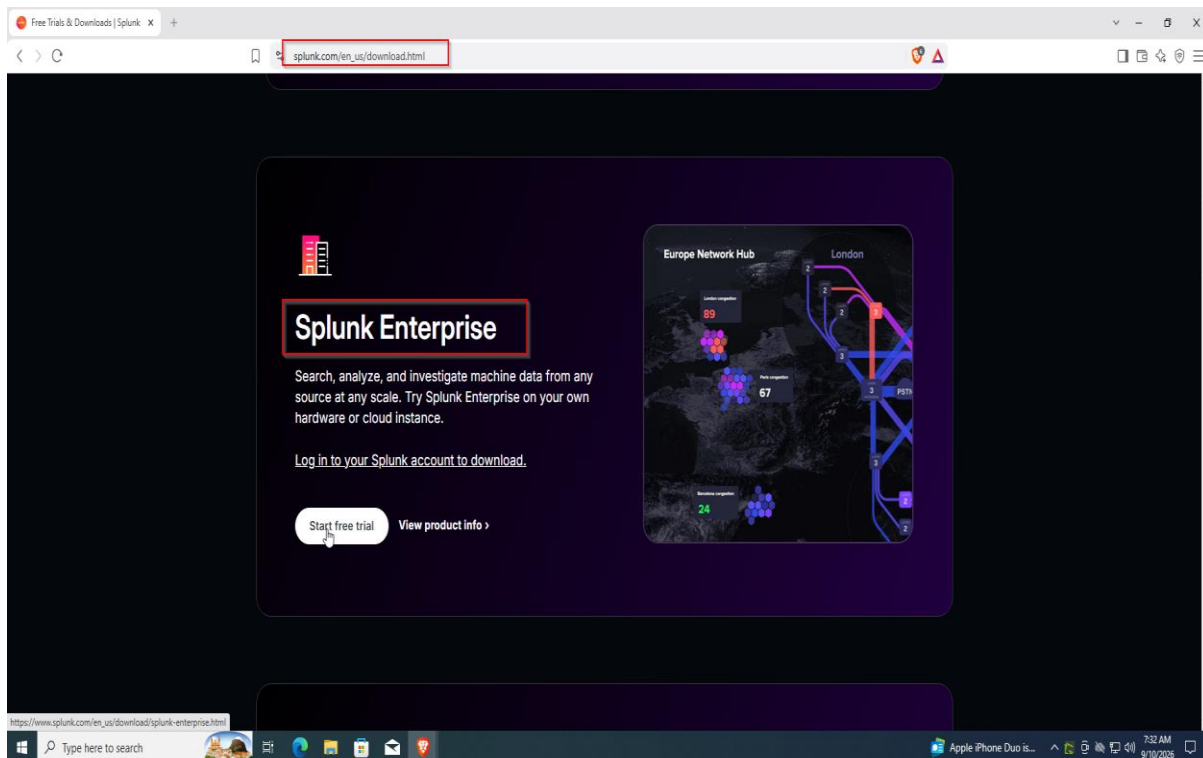
Windows Security Event ID 4625 - “An account failed to log on.”

What each value means:

Field	Your value	Meaning
FailureReason	%%2313	Windows internal message ID corresponding to unknown username or bad password in this context
SubStatus	0xc000006a	Bad/incorrect password ; the supplied account name was recognized, but authentication failed because of the password
LogonType	2	Interactive logon — someone/process attempted to log on locally at the computer console
LogonProcessName	User32	Windows User32 logon process handled the authentication request

Phase:4 – Splunk Setup

Step-7: First, search for Splunk and open the Splunk website. Scroll down to Splunk Enterprise, and then click Start Free Trial.



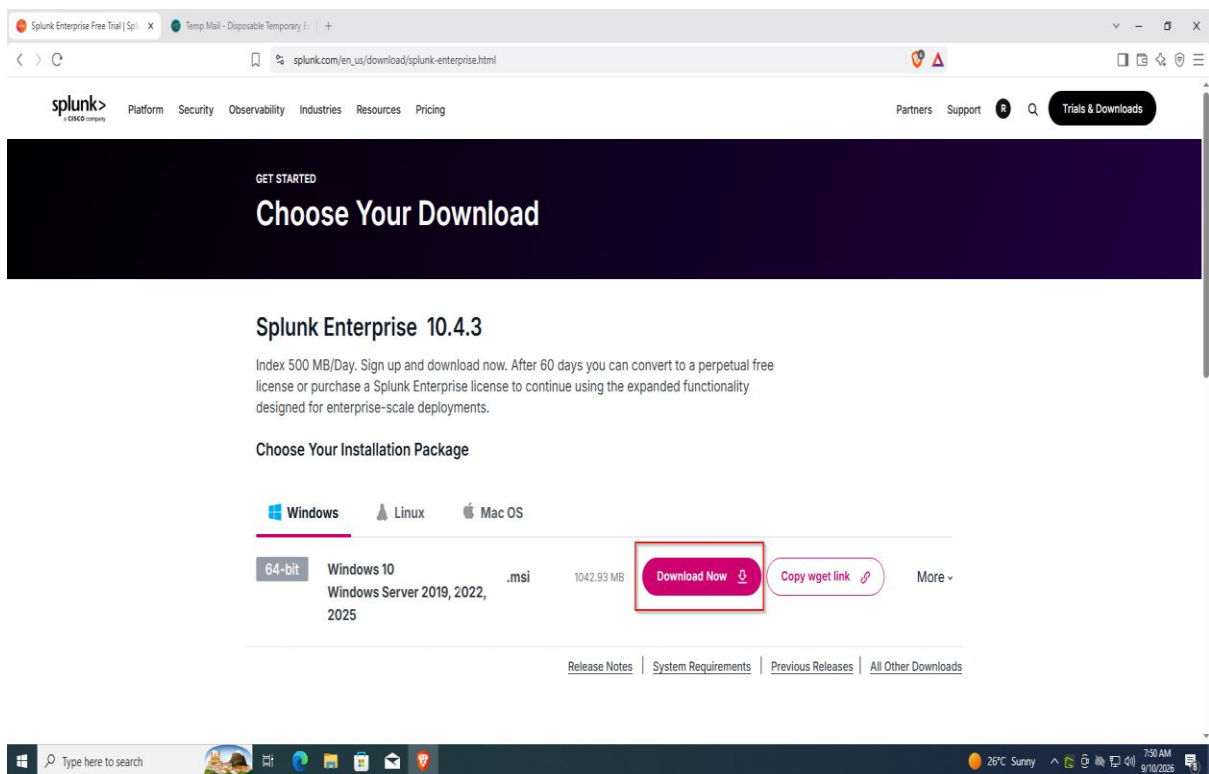
Step-8: Enter your information here and agree to the **terms and conditions**. Then, click **Create Your Account**.

The screenshot shows the Splunk Enterprise Free Trial registration page. The page has a dark purple background with white text. On the left, there are three dashboard preview images. Below them, text describes the benefits of the trial: 'Start searching, analyzing, and visualizing your data on powerful, easy-to-understand dashboards.' and 'Install on Windows or Linux to get insights from all parts of your multicloud and hybrid environment.' A list of benefits follows: 'Once you sign up for the Splunk Enterprise trial, you'll see how it helps you to:'. The benefits are: 'Tackle your hardest security and observability use cases.', 'Stream, collect, and index any data at any scale.', 'Set up real-time alerts so you can act fast.', 'Customize for your unique business needs with free, pre-built apps from Splunkbase.', and 'Administer your Splunk deployment on-premises or on your own cloud tenant.' Below this, there is a link to 'Try out Splunk Cloud Platform for free, or explore the rest of our free trials and downloads.' On the right, there is a registration form with the following fields: 'Job Title' (SOC), 'Phone Number' (8956471256), 'Company' (SOC), 'Country' (India), and 'Zip / Postal Code' (395201). Below the form, there are four checkboxes for agreeing to terms and conditions: 'I agree to the Splunk Website Terms & Conditions of Use and Splunk General Terms.', 'I agree to the Cisco's Online Privacy Statement.', 'Cisco may process my personal information, including transferring my personal information internationally, as described in its Online Privacy Statement.', and 'I agree to receive marketing communications by email, including educational materials, product and company announcements, and community event information, from Splunk Inc. and its subsidiaries pursuant to the terms of Cisco's Online Privacy Statement. I can unsubscribe at any time.' Below these checkboxes, there is a link to 'Visit here to learn more.' and a large pink 'Create your Account' button.

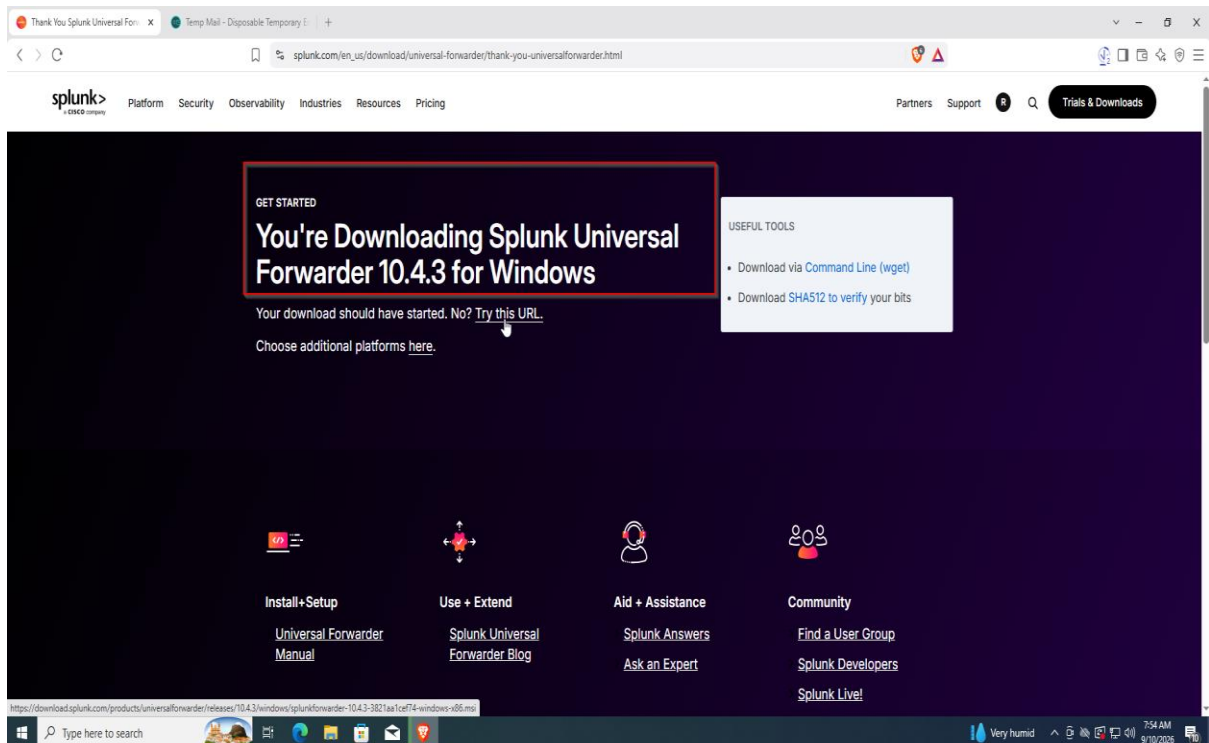
Step-9: Then, verify your account by entering the verification code and clicking Verify.

The screenshot shows the Splunk Enterprise 10.4.3 verification page. The page has a dark purple background with white text. At the top, there is a navigation bar with links to 'Platform', 'Security', 'Observability', 'Industries', 'Resources', and 'Pricing'. On the right, there are links to 'Partners', 'Support', and a 'Trials & Downloads' button. The main content area is divided into two sections. The left section is titled 'FREE TRIAL' and 'Splunk Enterprise 10.4.3'. It says 'Try Splunk Enterprise free for 60 days. No credit card required.' Below this, there is a dashboard preview titled 'Employee Device Security' showing various charts and maps. The right section is titled 'Verify your email'. It says 'We sent a 6-digit code to j***9@94an.com. Please enter it here to verify your email.' Below this, there is a text input field containing the code '280432' and a large pink 'Verify' button.

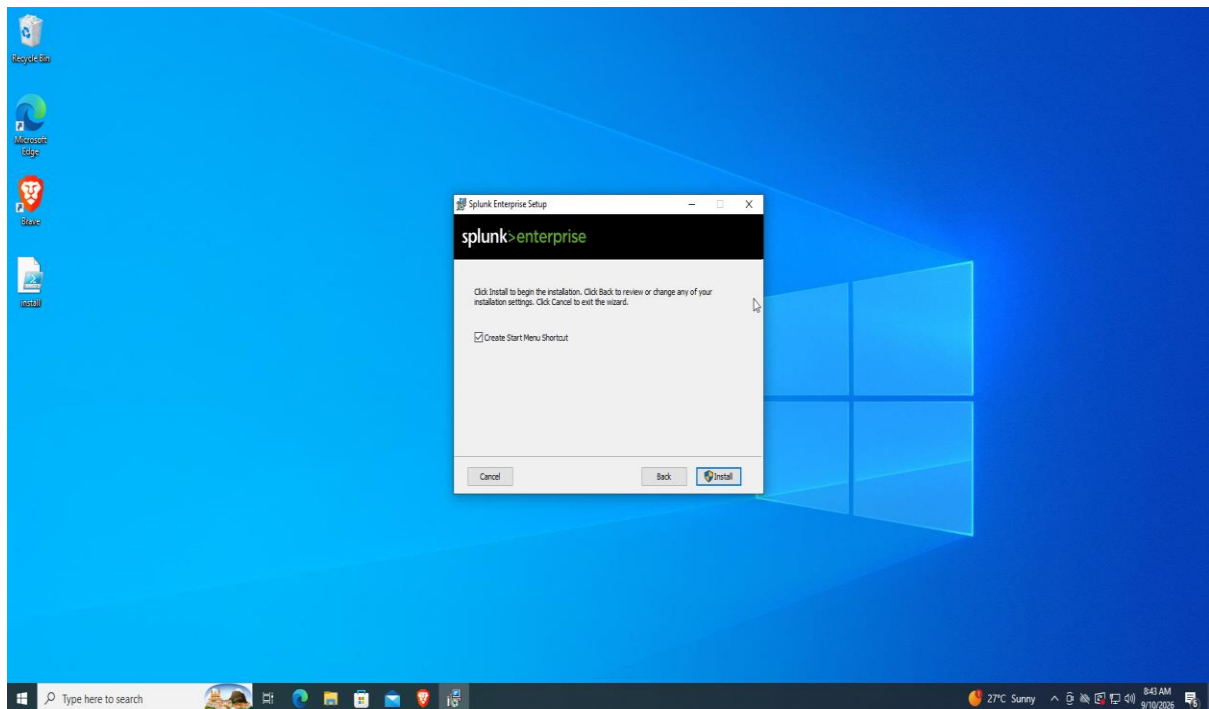
Step-10: Here, you can see the Installation Package. Click Download.



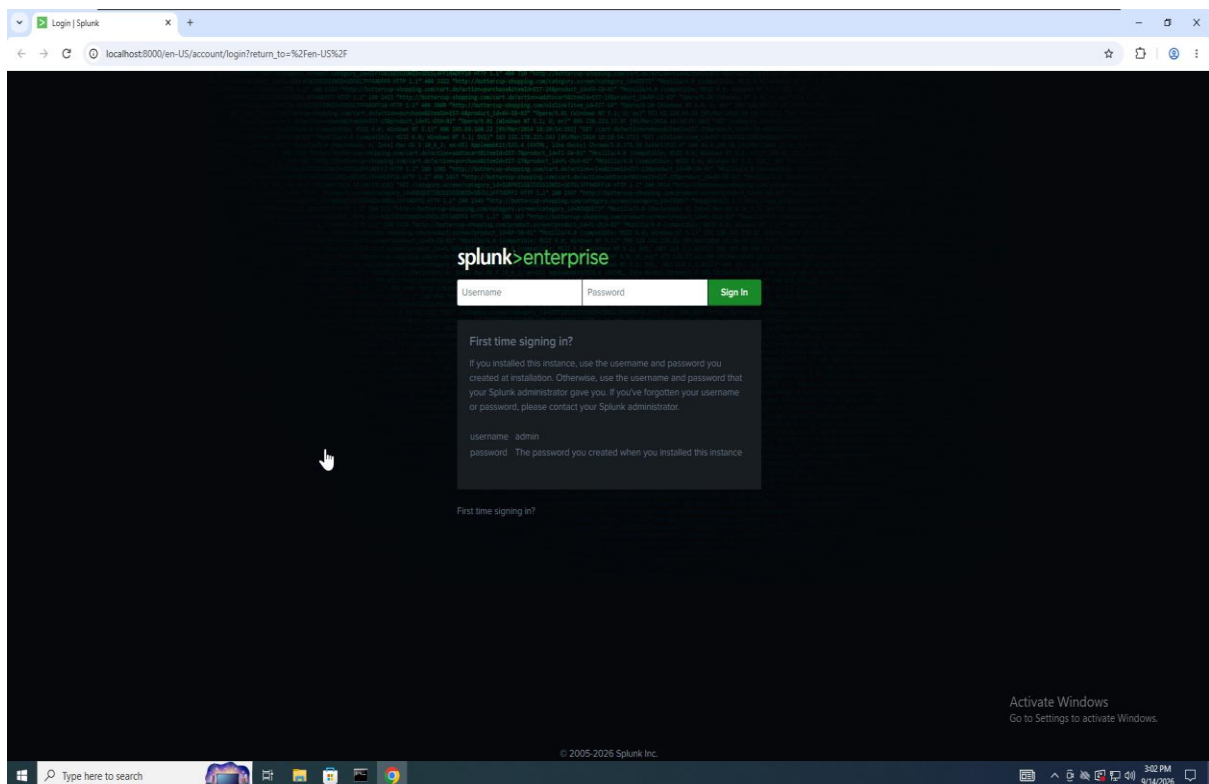
Step-11: Log in to your Splunk account again and download the Splunk Universal Forwarder. Then, click Try This URL.



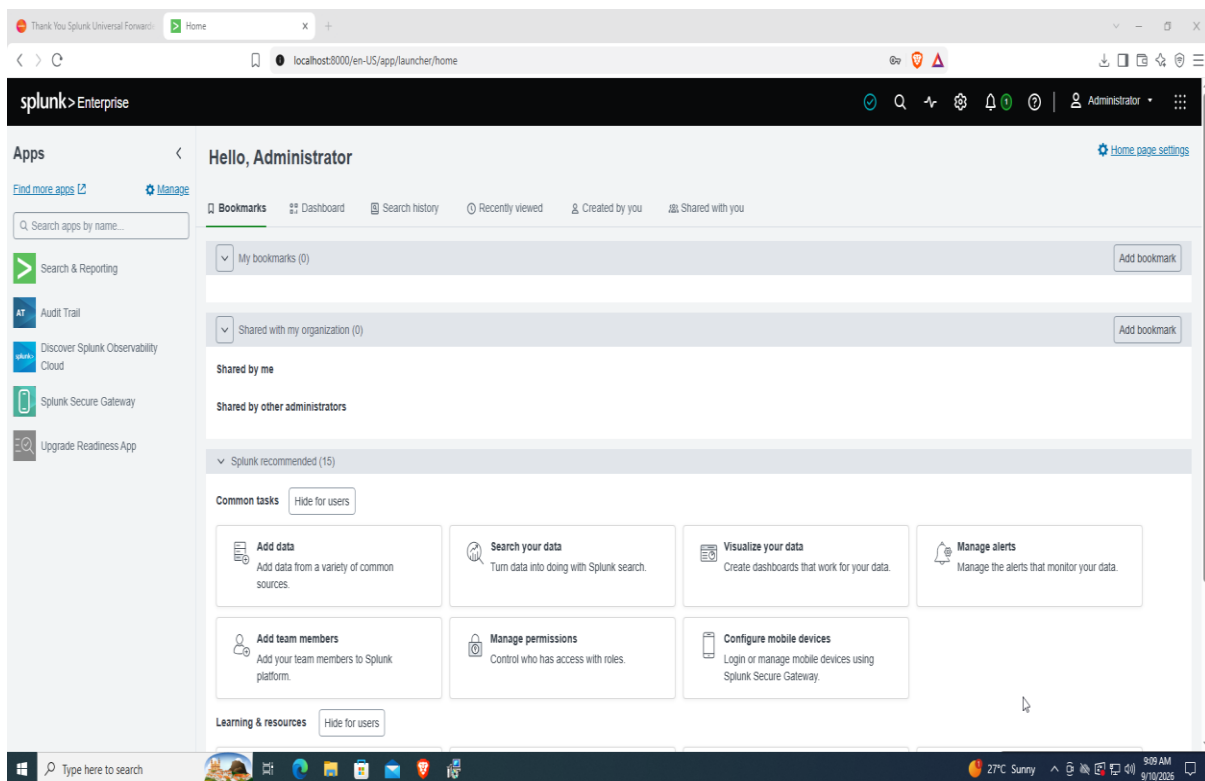
Step-12: Click Install to begin the installation. Once the installation is complete, the Splunk Enterprise Setup will be ready.



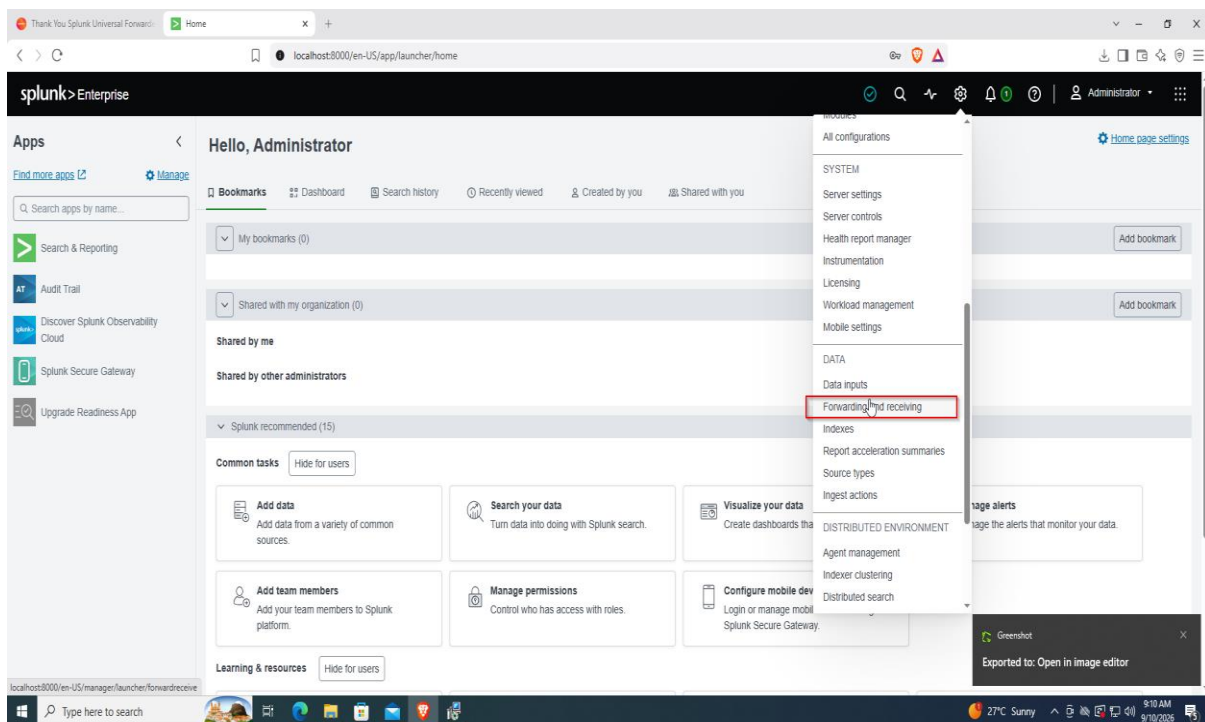
Step-13: Here, you can see the Splunk interface.



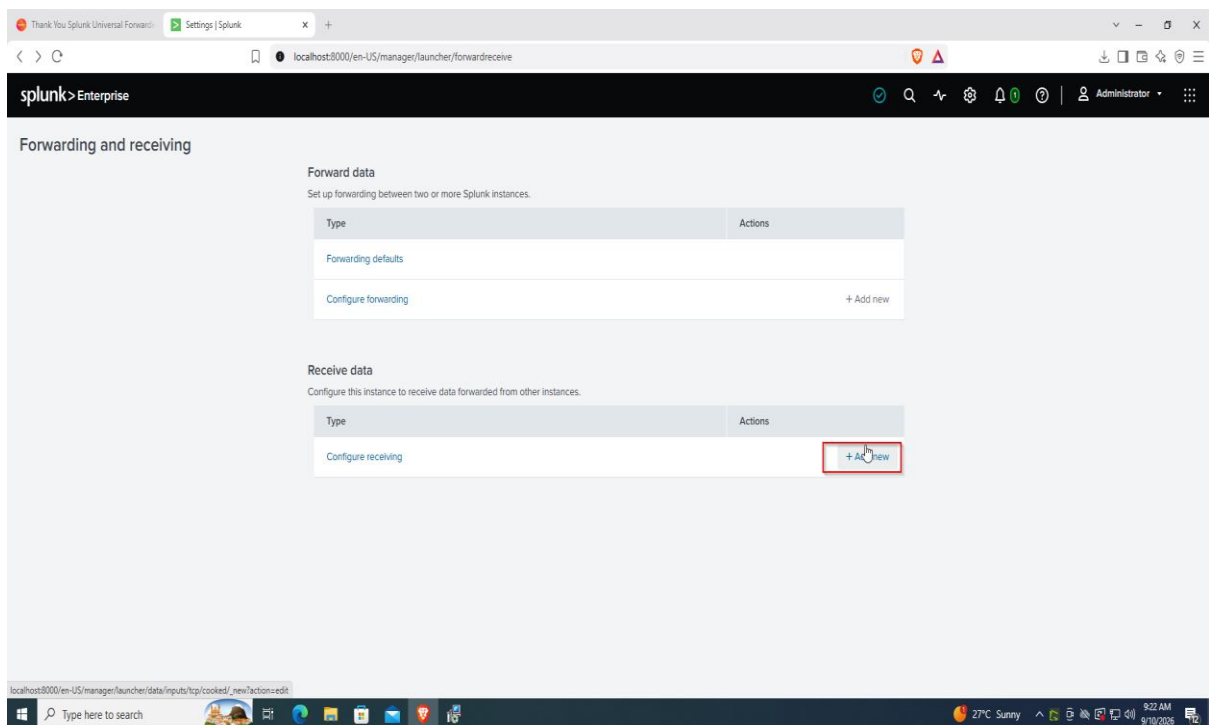
Step-14: Here, you can see Administrator of Splunk Enterprise Interface.



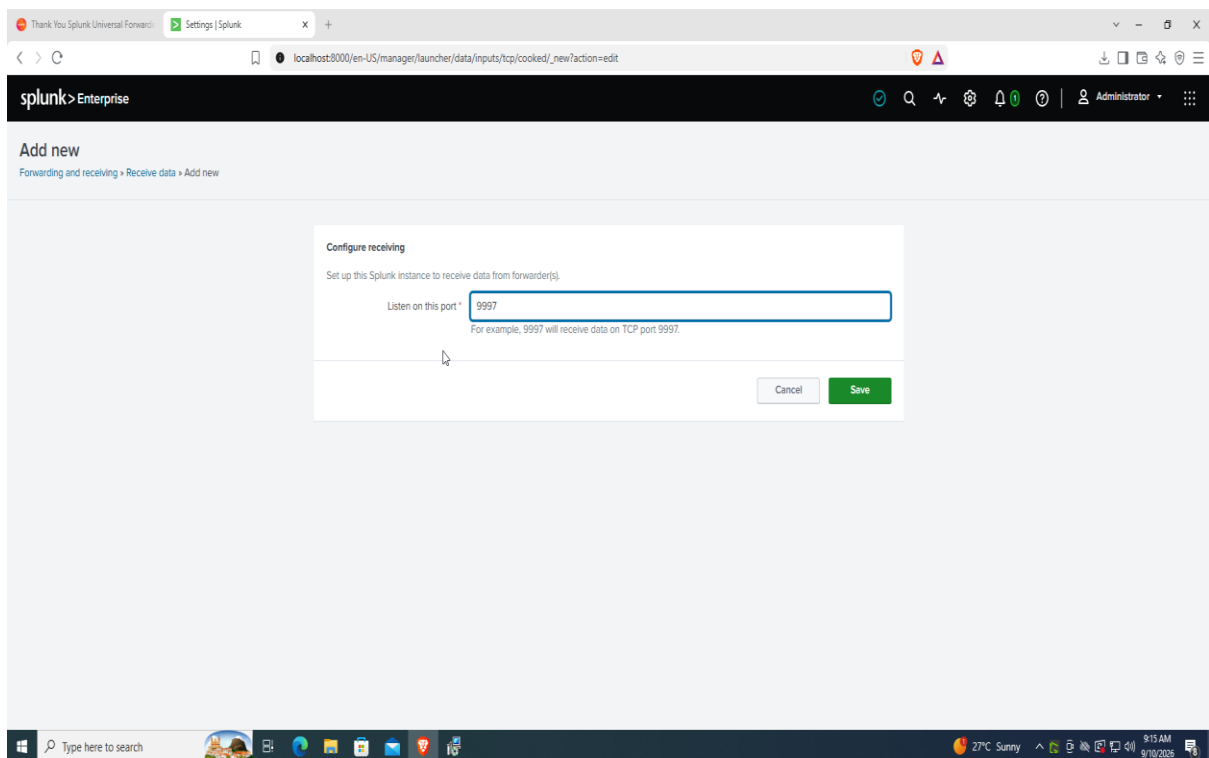
Step-15: Click Settings, then select Forwarding and Receiving.



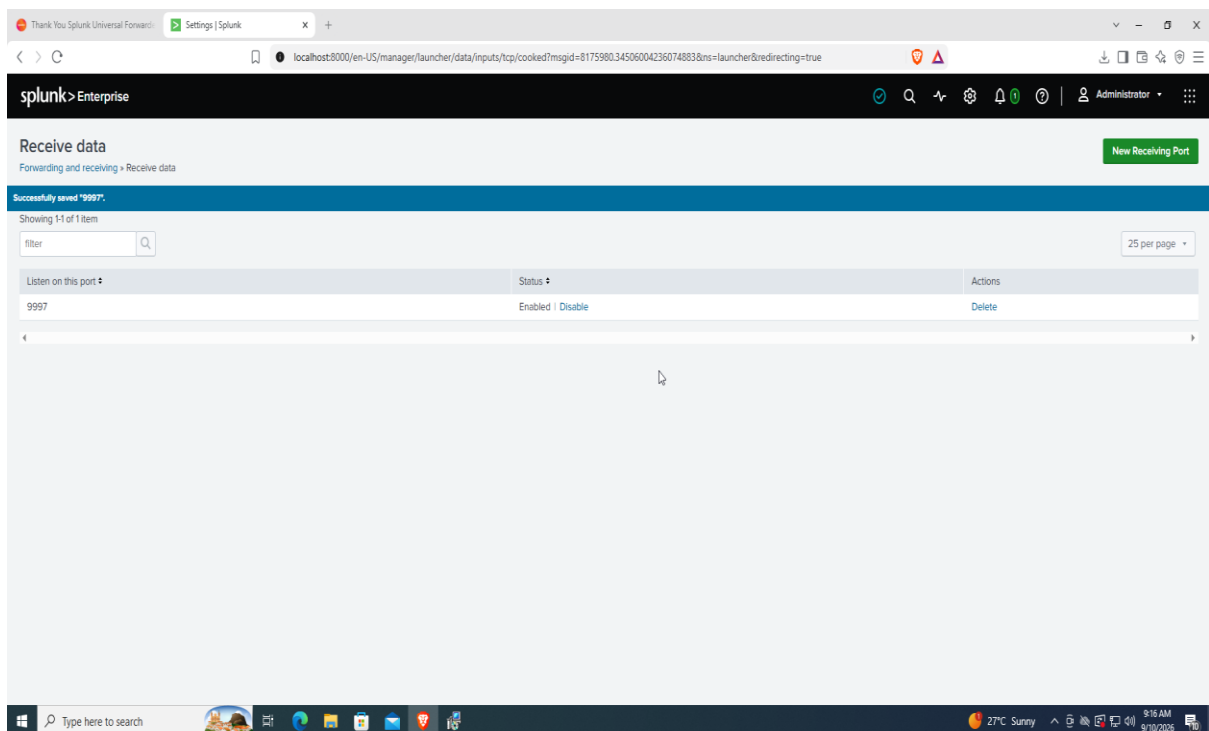
Step-16: Here, you can configure data receiving. Click the Add New button.



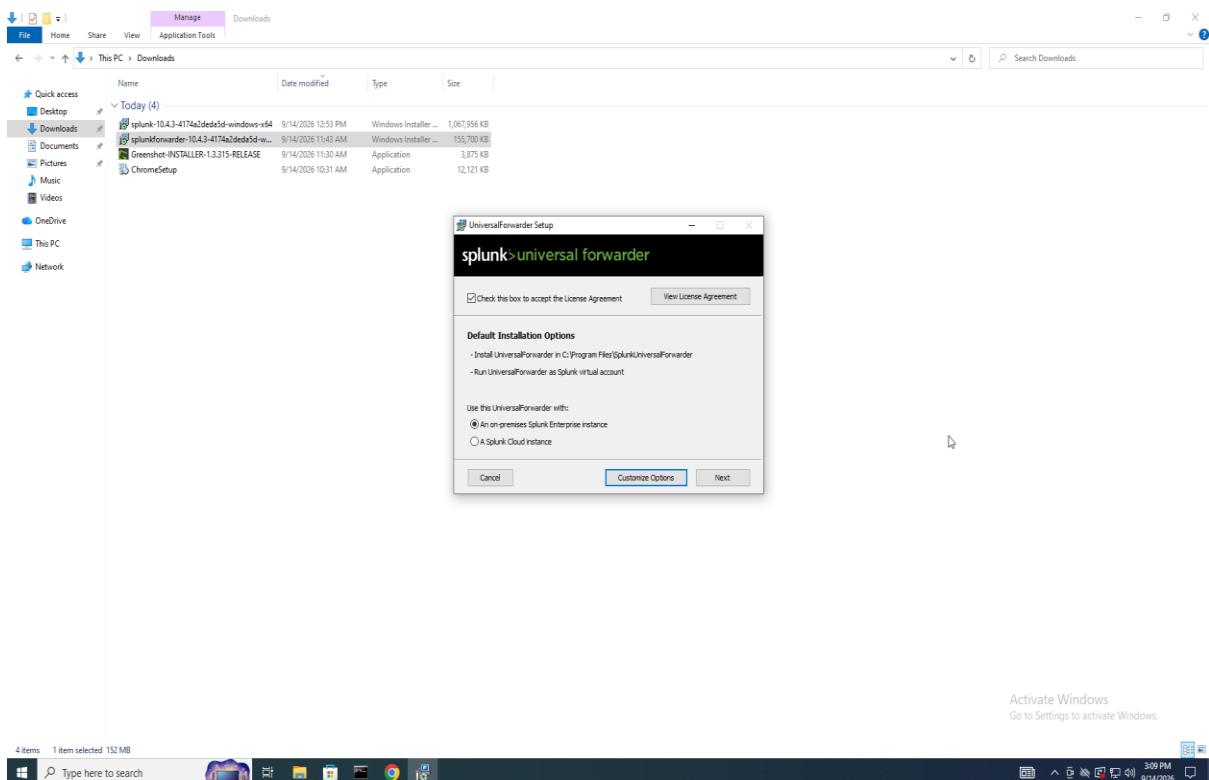
Step-17: Set the Listen on this port value to 9997, then click Save.



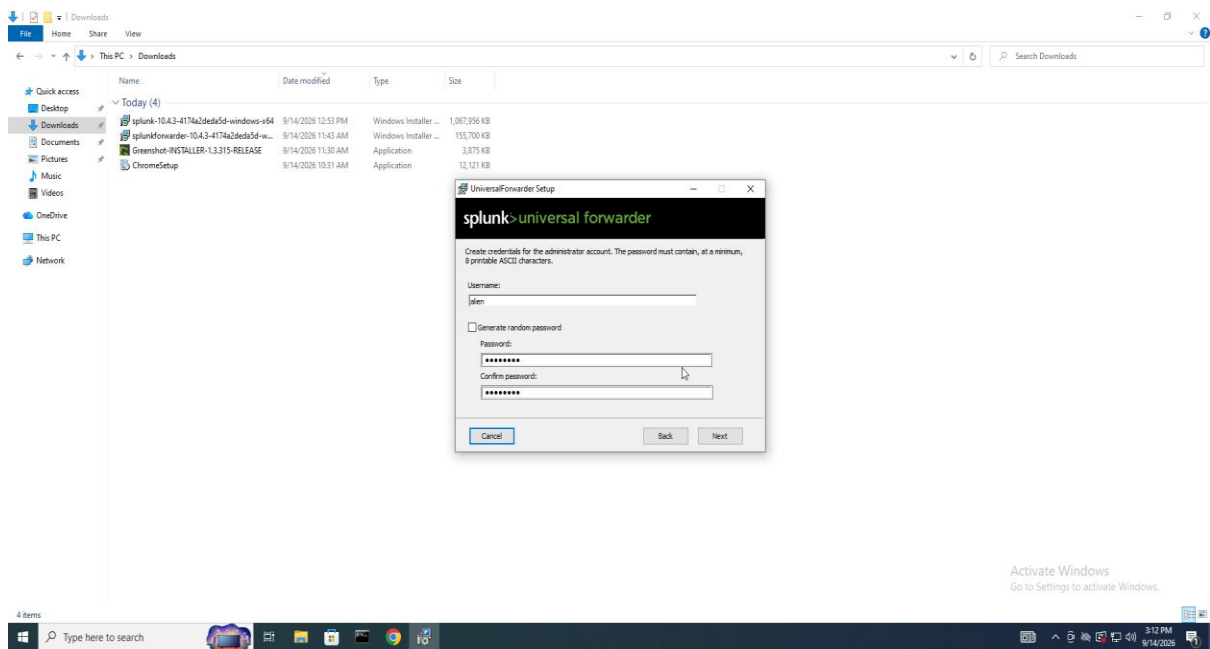
Step-18: Here, you can see that the listening port has been successfully configured.



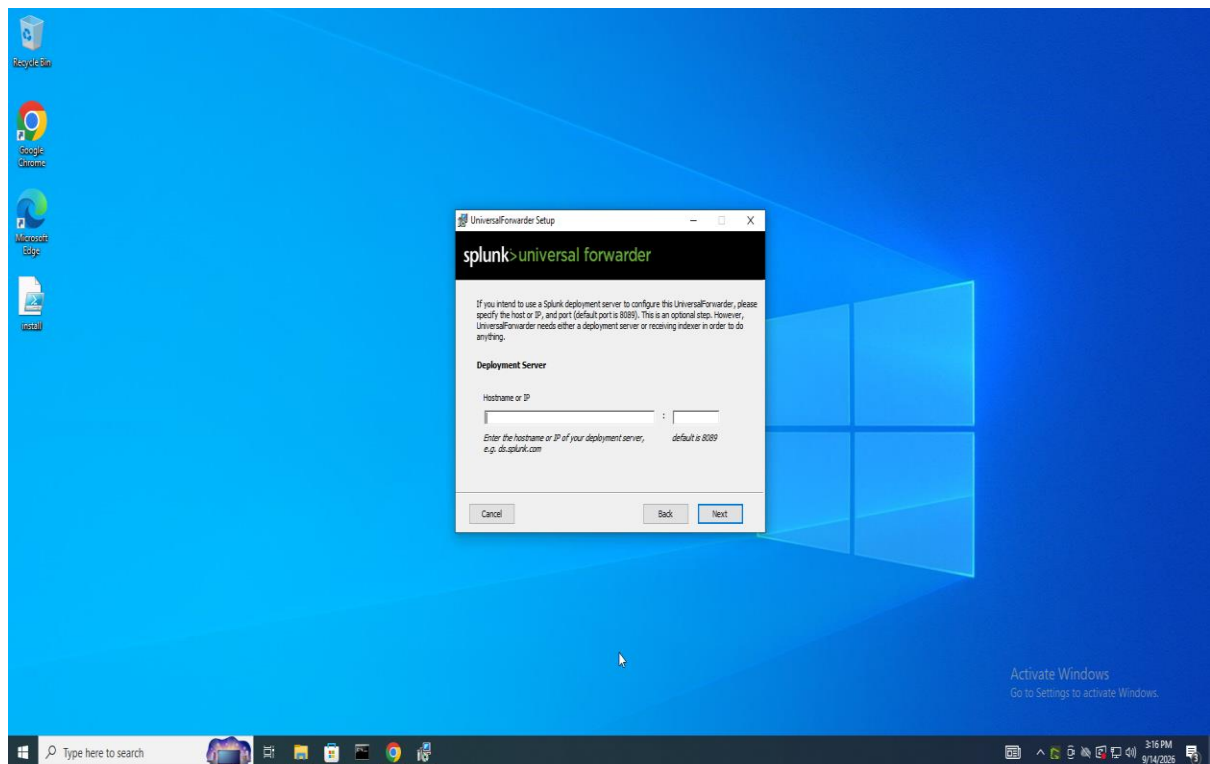
Step-19: Now, install **Splunk Universal Forwarder** and click next.



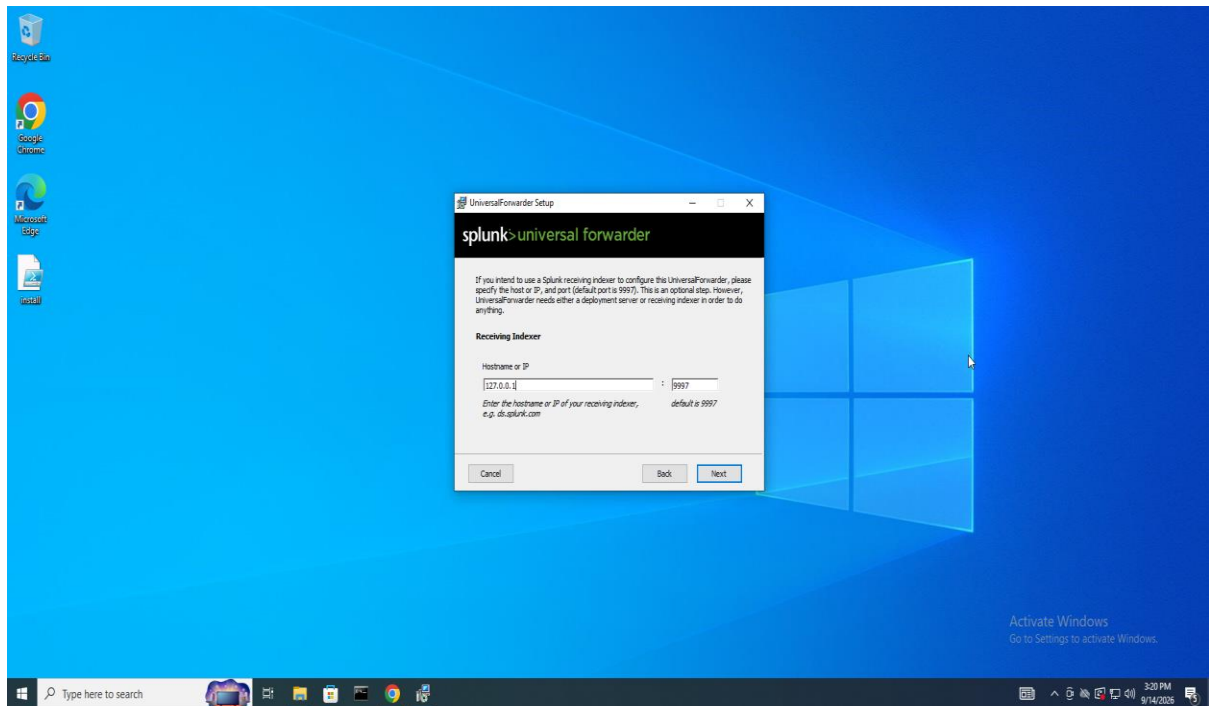
Step-20: Here, create the credentials for the administrator account and click Next.



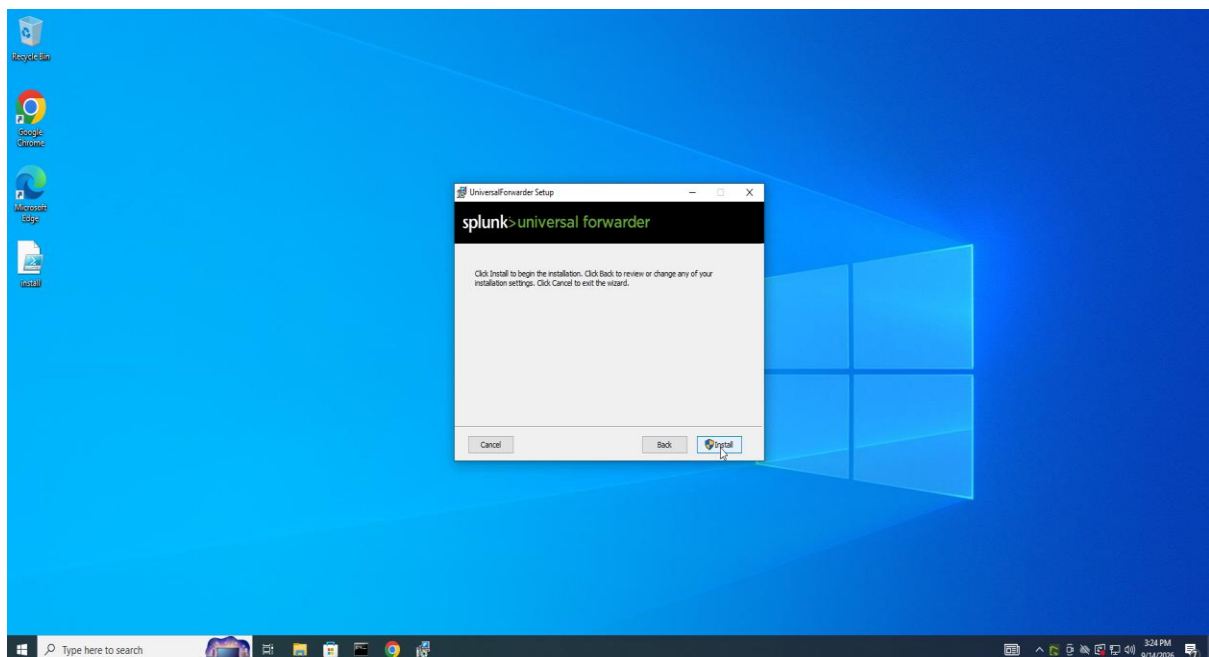
Step-21: you can leave the Deployment Server fields blank and click Next.



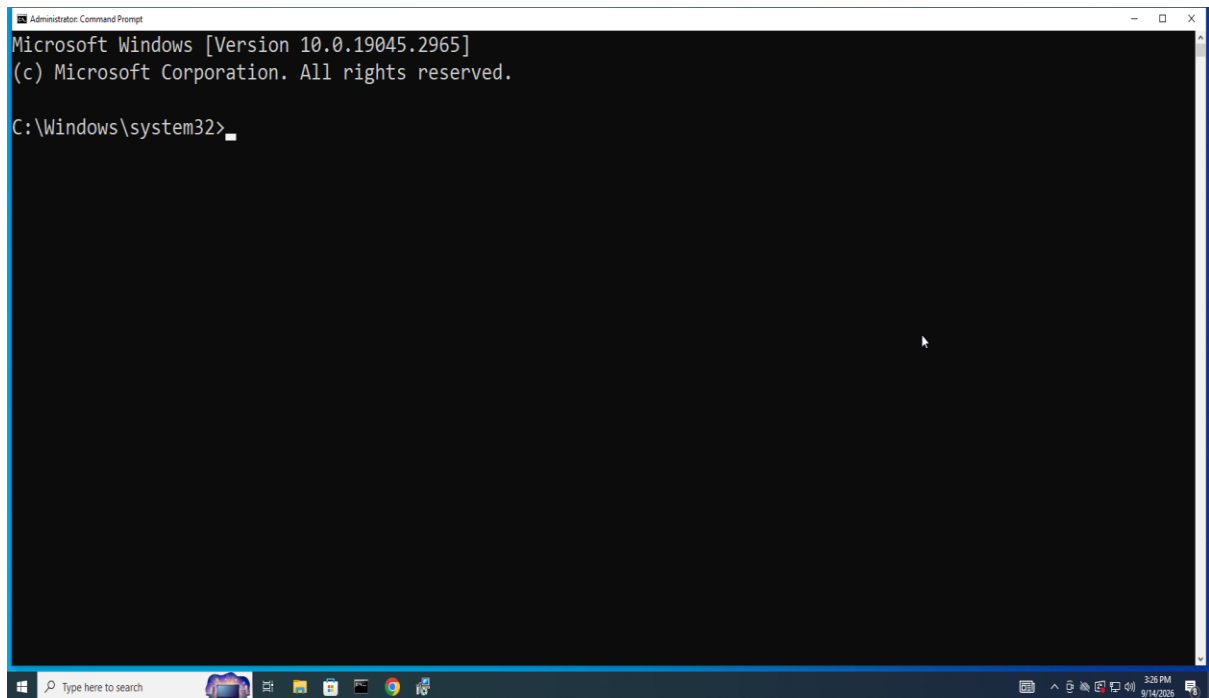
Step-22: Here, if **Splunk Enterprise** is installed on the same laptop, we can use **127.0.0.1:9997**. This is the forwarder's output/receiving configuration, which we'll configure separately. **Click Next.**



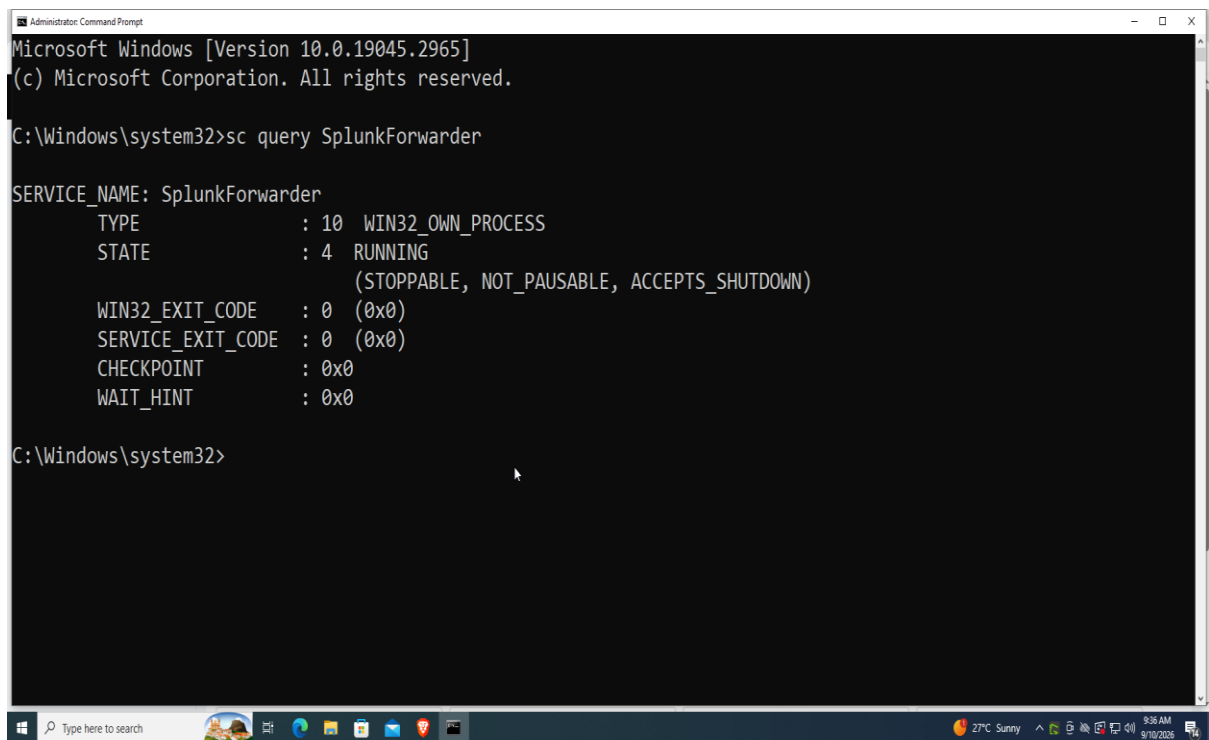
Step-23: Last step: Click the Install button to begin the installation.



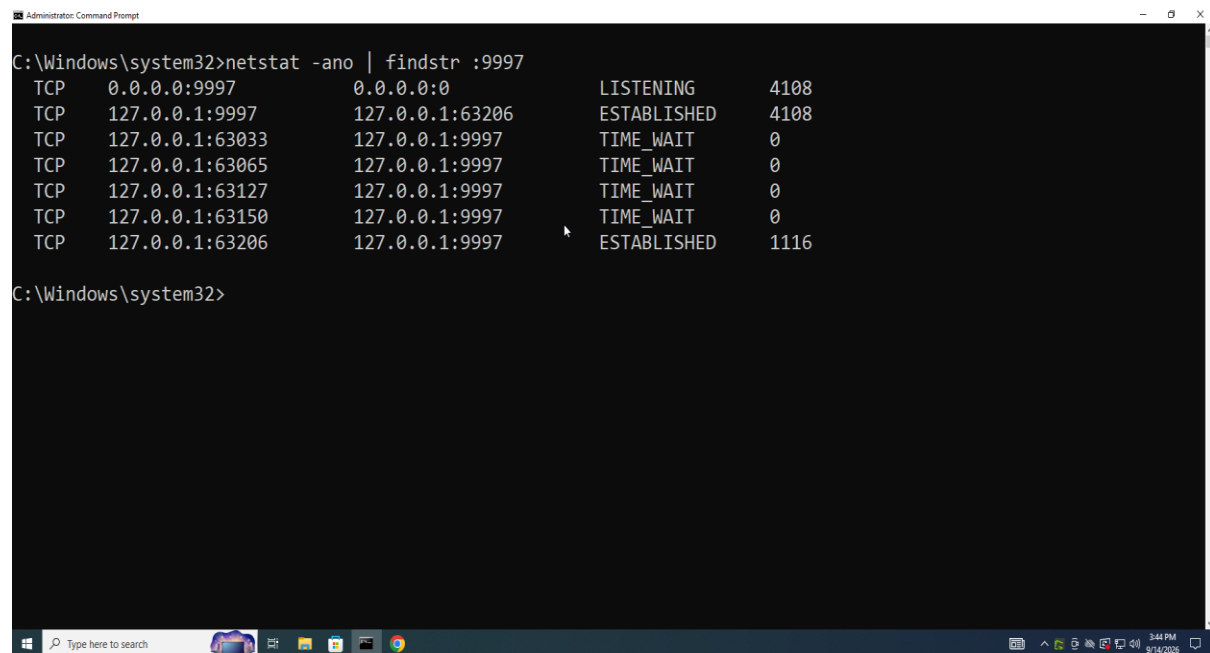
Step-24: Once the **Splunk Universal Forwarder** installation is complete, click **Finish**. Then, open **Command Prompt as Administrator**.



Step-25: Here, you can verify that the Splunk Universal Forwarder service is running successfully, as indicated by the **RUNNING** status.



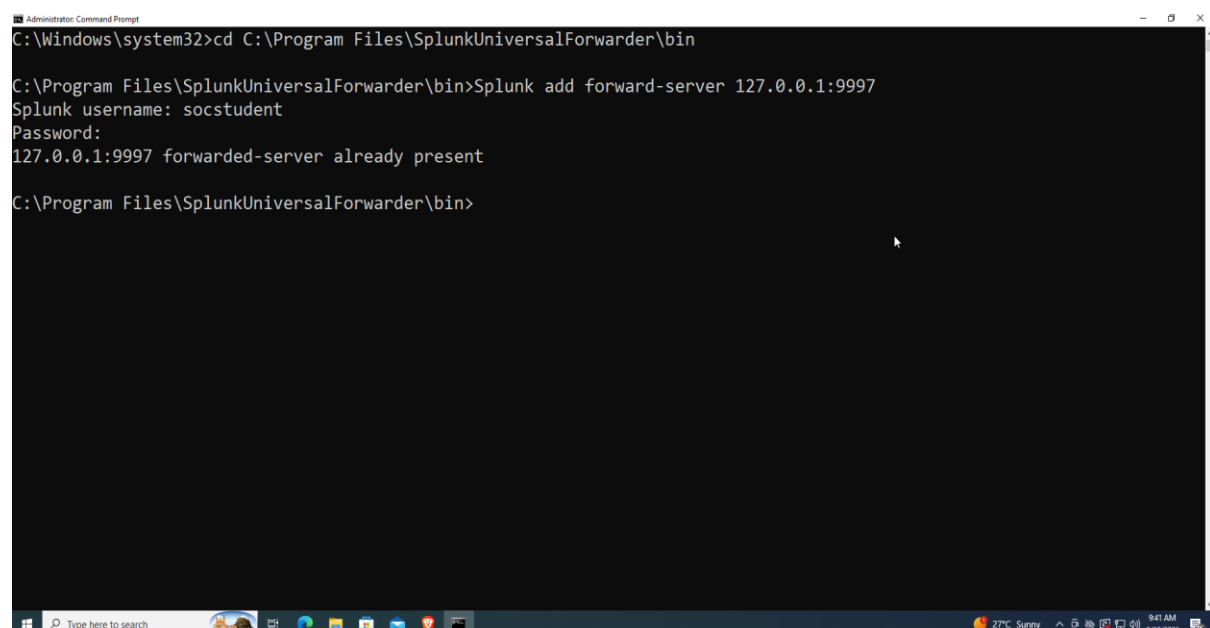
Step-26: 0.0.0.0:9997 means it is not restricted to only 127.0.0.1. It is listening on port 9997 across the machine's network interfaces.



```
Administrator: Command Prompt
C:\Windows\system32>netstat -ano | findstr :9997
TCP    0.0.0.0:9997        0.0.0.0:0          LISTENING          4108
TCP    127.0.0.1:9997      127.0.0.1:63206    ESTABLISHED        4108
TCP    127.0.0.1:63033     127.0.0.1:9997     TIME_WAIT           0
TCP    127.0.0.1:63065     127.0.0.1:9997     TIME_WAIT           0
TCP    127.0.0.1:63127     127.0.0.1:9997     TIME_WAIT           0
TCP    127.0.0.1:63150     127.0.0.1:9997     TIME_WAIT           0
TCP    127.0.0.1:63206     127.0.0.1:9997     ESTABLISHED        1116
C:\Windows\system32>
```

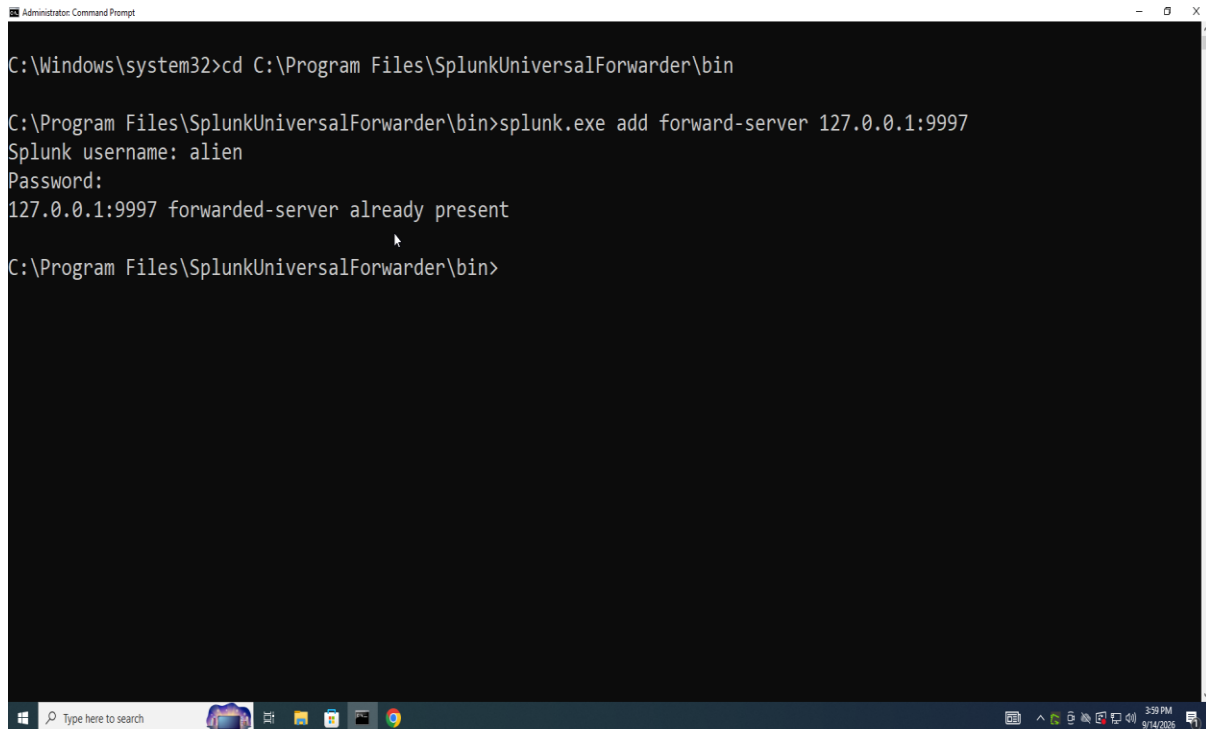
Step-27: Here, you can see that the Splunk Universal Forwarder is already configured with the forwarding server 127.0.0.1:9997. The message “forwarded server already present” confirms that the forwarding server has already been added.

Note: Each Universal Forwarder can send logs to the same Splunk server IP and receiving port. The Splunk server can then collect and analyze events from multiple Windows endpoints.



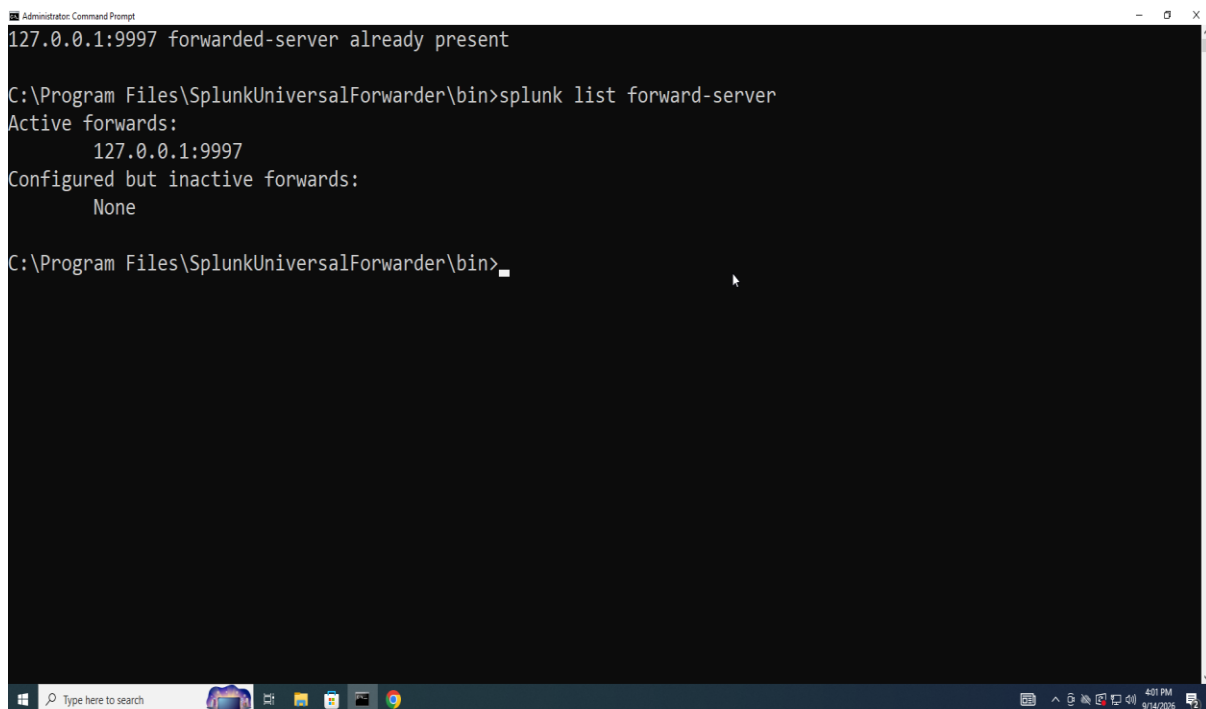
```
Administrator: Command Prompt
C:\Windows\system32>cd C:\Program Files\SplunkUniversalForwarder\bin
C:\Program Files\SplunkUniversalForwarder\bin>Splunk add forward-server 127.0.0.1:9997
Splunk username: socstudent
Password:
127.0.0.1:9997 forwarded-server already present
C:\Program Files\SplunkUniversalForwarder\bin>
```

Step-28: Change the directory help of cd command giving path of SplunkUniversalForwarder. Using Splunk add forwarder-server 127.0.0.1:9997. Enter Splunk username and password. Here, you can status forwarder-server already present.



```
Administrator: Command Prompt
C:\Windows\system32>cd C:\Program Files\SplunkUniversalForwarder\bin
C:\Program Files\SplunkUniversalForwarder\bin>splunk.exe add forward-server 127.0.0.1:9997
Splunk username: alien
Password:
127.0.0.1:9997 forwarded-server already present
C:\Program Files\SplunkUniversalForwarder\bin>
```

Step-29: Splunk list forward-server using this command showing that Active forward status on hostname 127.0.0.1 and ID 9997.

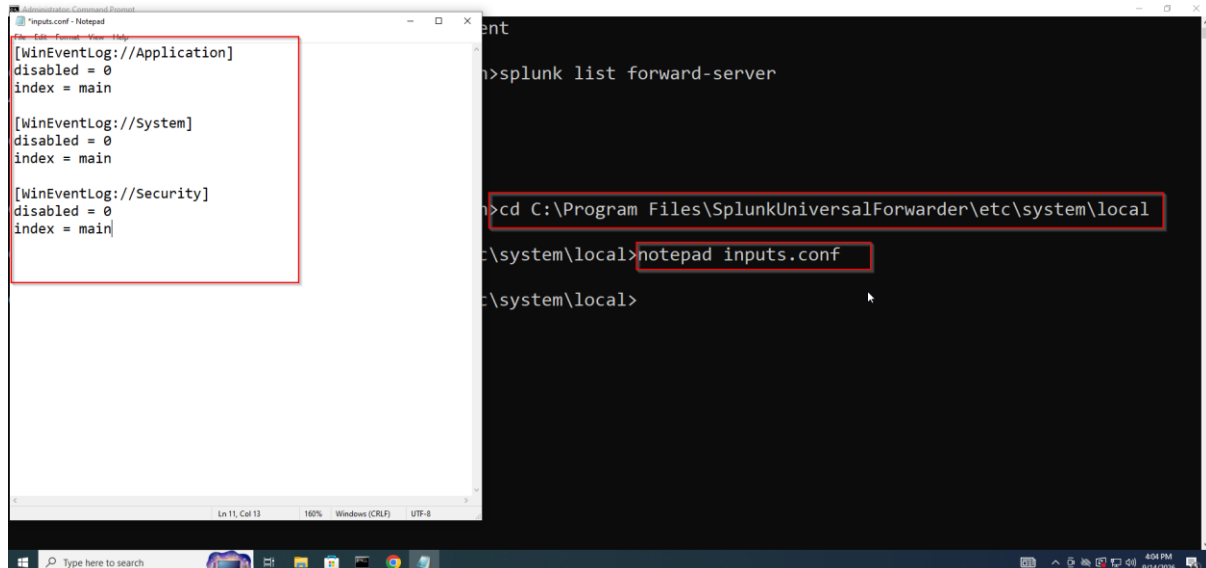


```
Administrator: Command Prompt
127.0.0.1:9997 forwarded-server already present
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Active forwards:
    127.0.0.1:9997
Configured but inactive forwards:
    None
C:\Program Files\SplunkUniversalForwarder\bin>
```

Step-30: Now, navigate to the **local configuration folder** and run the following command:

Command: notepad inputs.conf

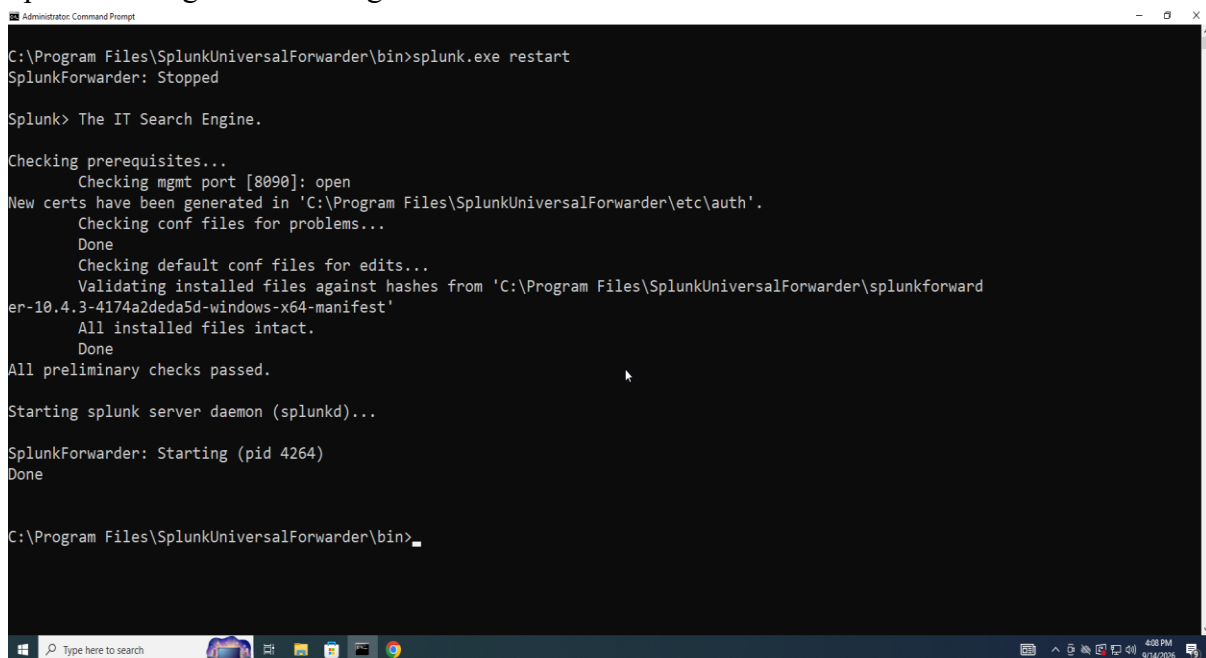
Working: This command creates or opens the inputs.conf configuration file in Notepad. The file is used to configure and define the data sources that the **Splunk Universal Forwarder** will collect and forward to the Splunk server.



Step-31: Now, open **Command Prompt as Administrator** and run the following command:

Command: splunk.exe restart

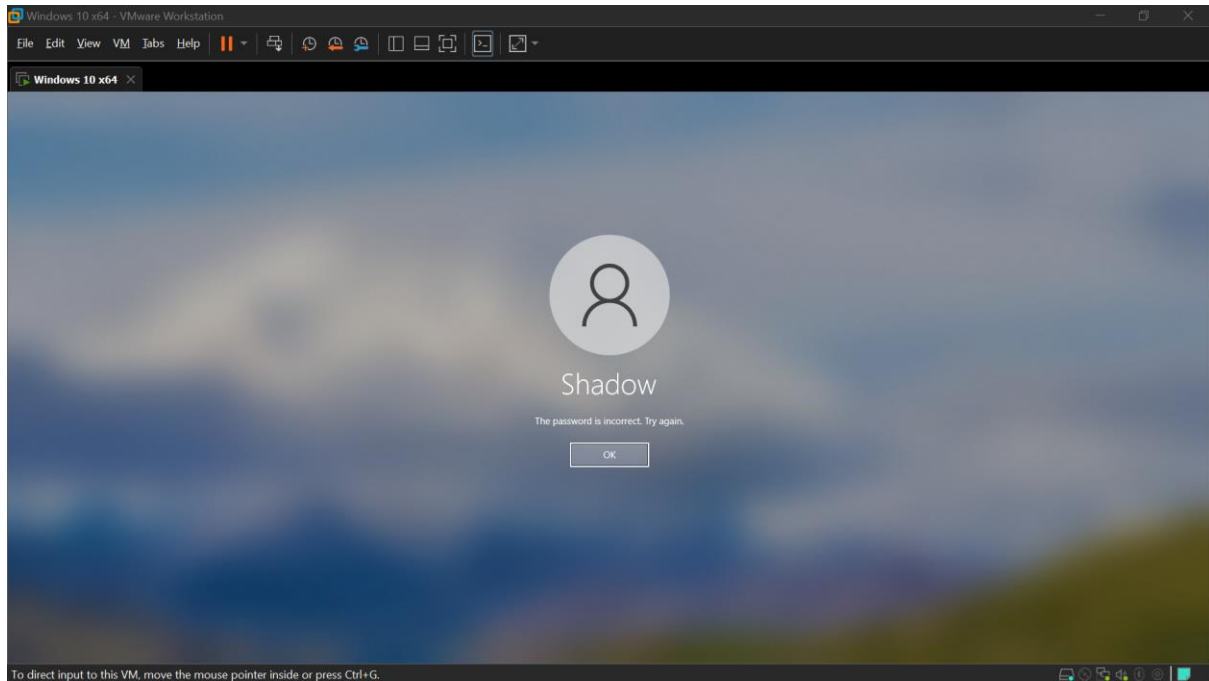
Working: This command restarts the **Splunk Universal Forwarder** and applies the newly updated configuration settings.



Successfully completed the full setup and configuration of Splunk Enterprise and Splunk Universal Forwarder.

Phase-5: Detection-1. Failed login

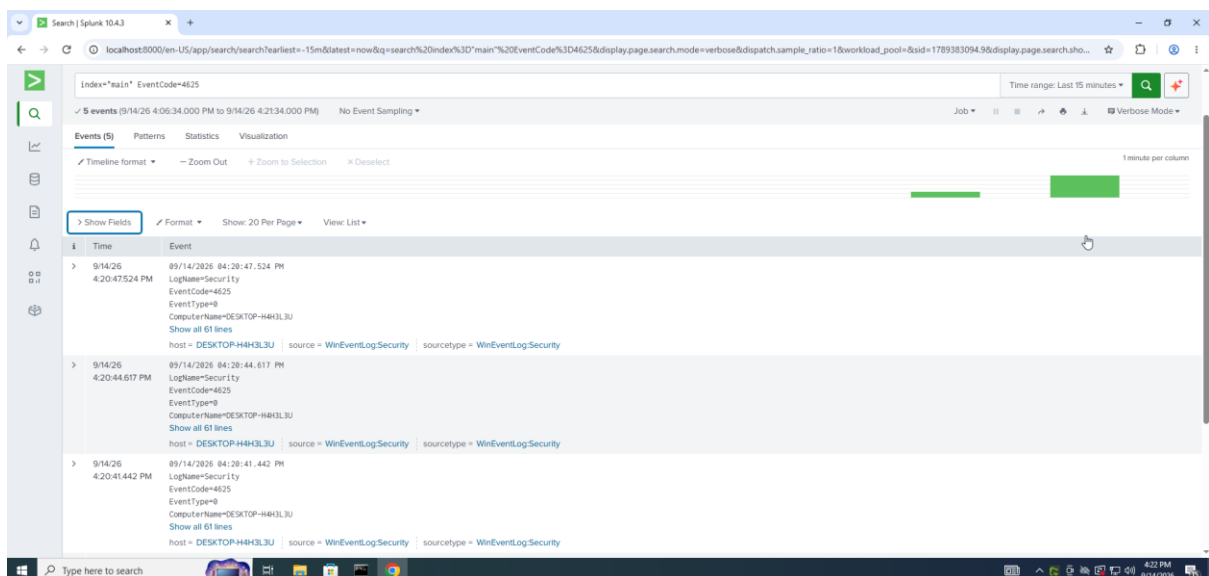
Step-32: Here, entering wrong password.

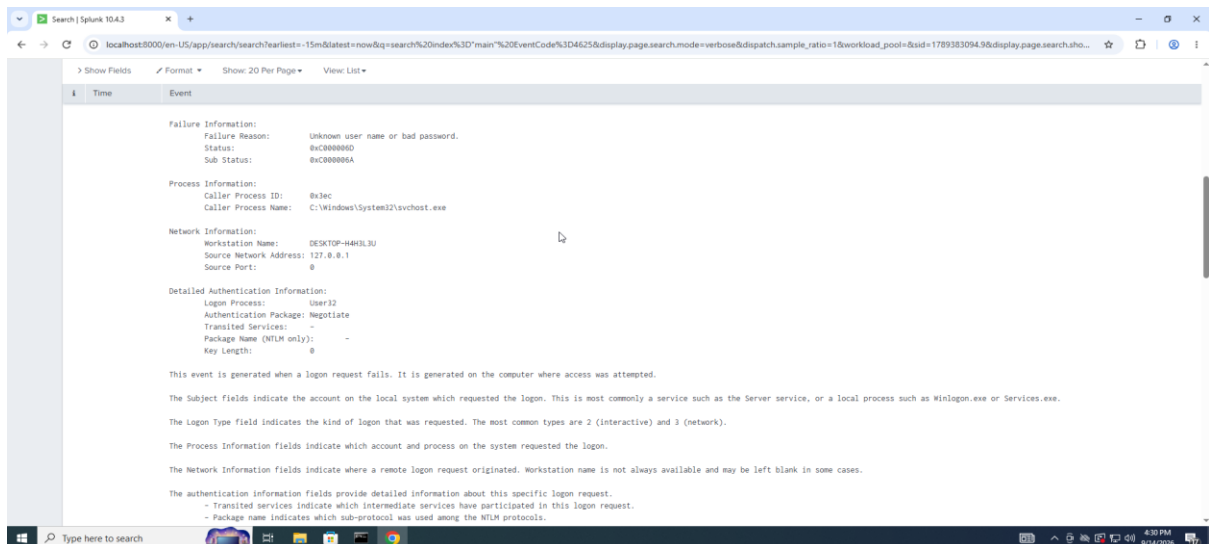
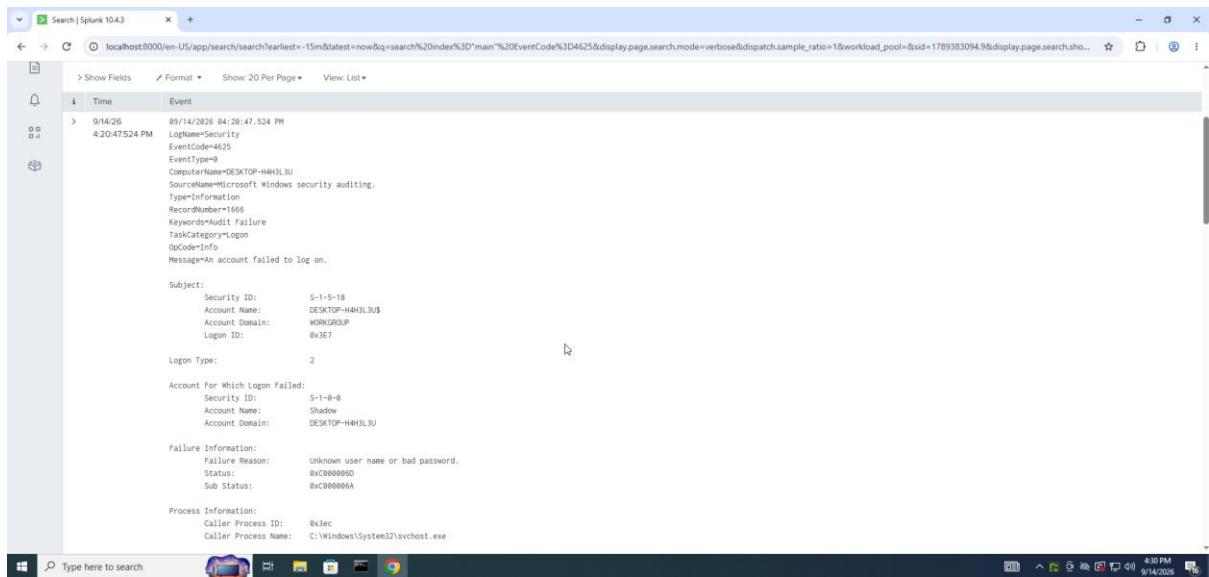


Step-33: We can search for failed authentication events with SPL along the lines of:

Query: index="main" EventCode=4625

Working: This Splunk query searches the main index for Windows Security Event ID 4625, which indicates a failed logon attempt. It can be used to identify unsuccessful authentication attempts and investigate potential unauthorized access or brute-force activity.





Then we can investigate fields such as:

Account Name: Shadow

Computer Name: DESKTOP-H4H3L3U

Source Network Address: 127.0.0.1

Task Category =Logon

Logon Process: User32

Failure Reason: Unknown username or bad password.

_time: 04:20:44

Note: The exact field names depend on how your Windows logs are being ingested, so we won't blindly copy a query. We'll first inspect your actual Splunk data.

Step-34: Turn it into a SOC detection:

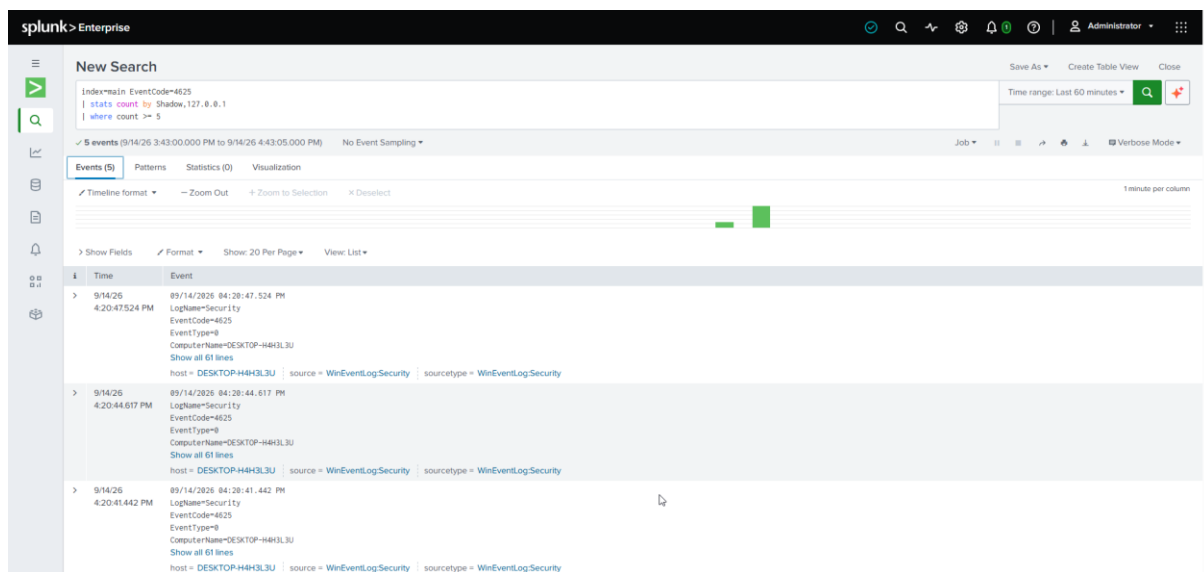
Once we confirm the event works, we'll make a query that detects repeated failures.

Command: index=main EventCode=4625

| stats count by Shadow,127.0.0.1

| where count >= 5

Working: The query is intended to help detect multiple failed login attempts, which may indicate brute-force activity or suspicious authentication behavior.



Meaning:

Failed login

Failed login

Failed login

Failed login

Failed login

↓

5 attempts

↓

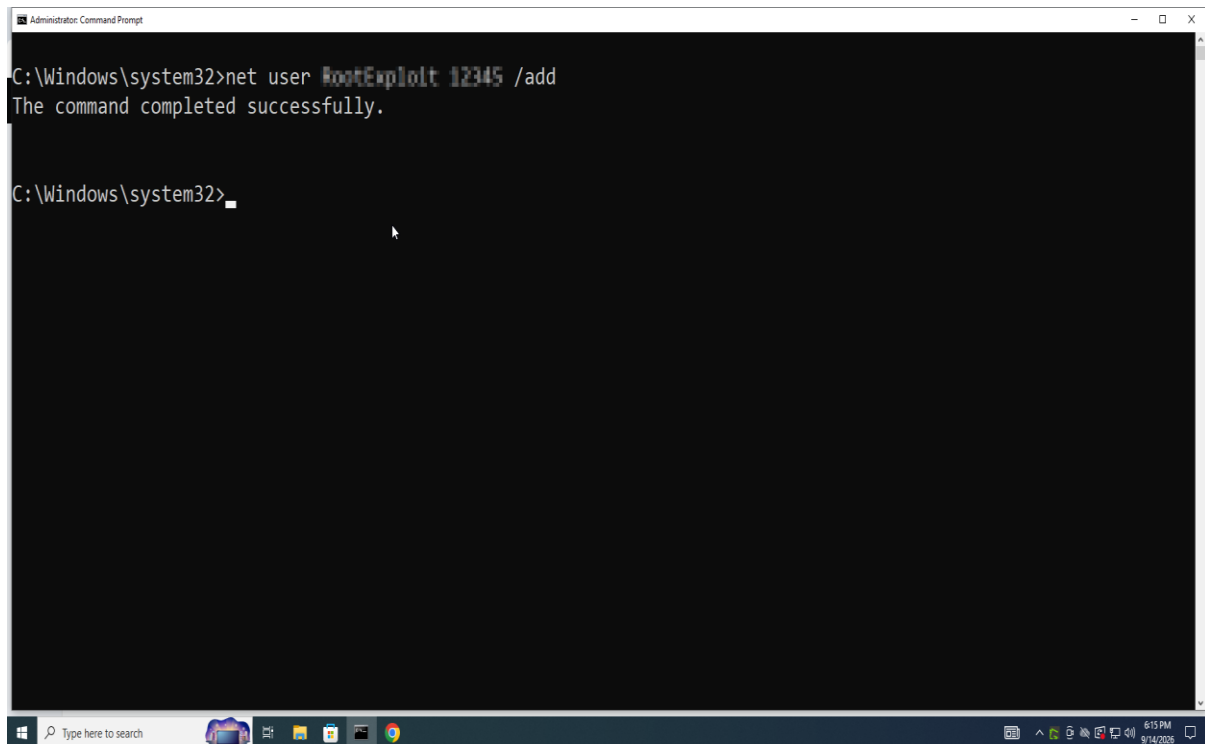
Potential brute-force activity

↓

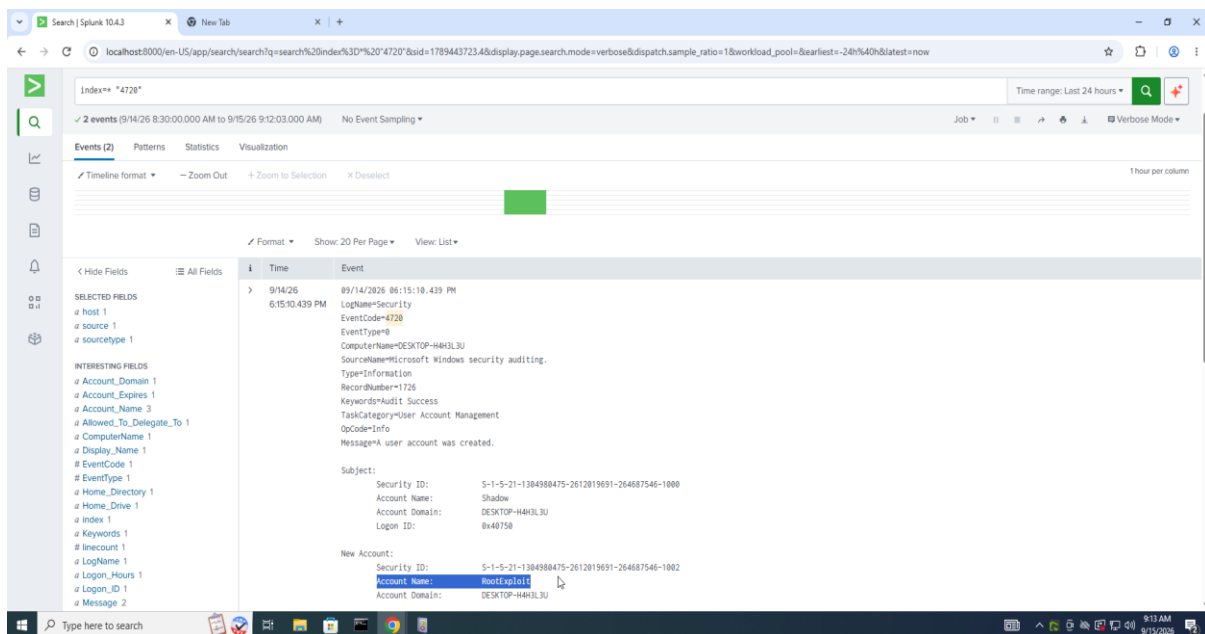
SOC Alert

Phase-6: Detection-2. New user

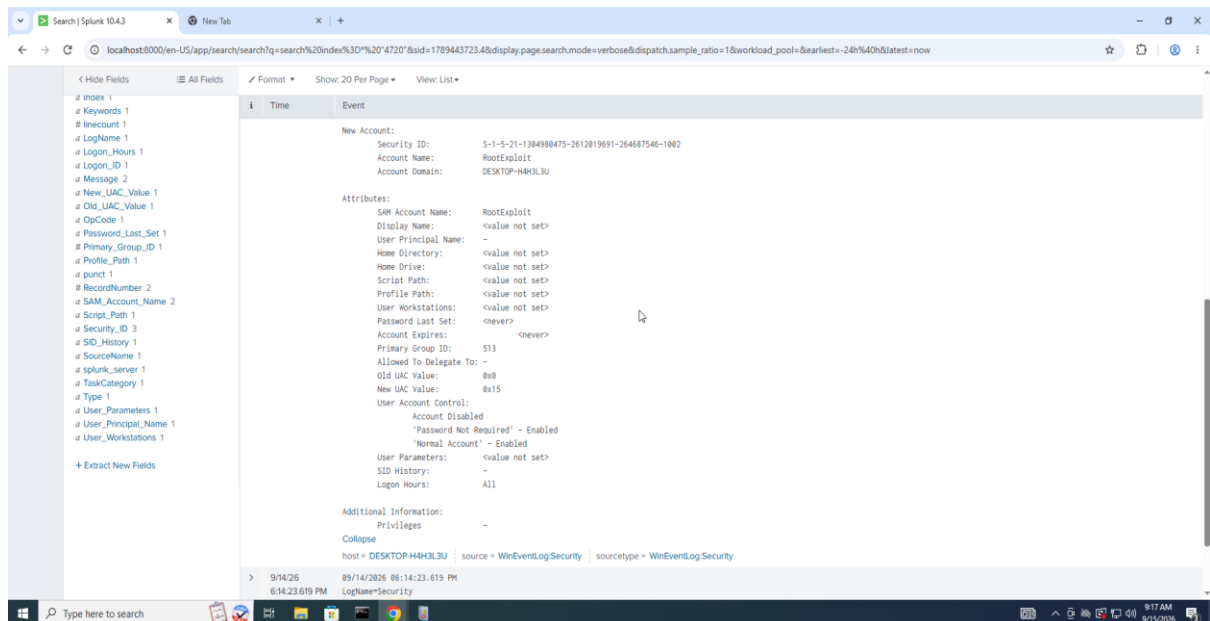
Step-35: We'll generate a test account in your own Windows environment.



Step-36: The same 4720 event was received and parsed, showing details such as RootExploit, the creator Shadow, computer name, SID, and account settings.



Step-37: An analyst can use these fields to determine who created the account, when, where, and whether the new account is suspicious.



1.What can a SOC analyst see?

For Event ID 4720, you can investigate several things:

Field	What it tells you
EventCode	Windows event type — 4720
_time	When the account was created
ComputerName	Which Windows machine
Account_Name	Parsed account name; inspect event context
Account_Domain	Domain/local computer associated with account
Subject	Account that performed the action
New Account	Account that was created
Security ID / SID	Unique Windows identifier
Logon_ID	Links the action to a Windows logon session
Message	Human-readable event description
TaskCategory	User Account Management

Field	What it tells you
Keywords	Audit Success

2. The most important SOC question:

Ask:

WHO created the account, WHAT account was created, WHERE was it created, and WHEN?

WHO?

Shadow

WHAT?

RootExploit

WHERE?

DESKTOP-H4H3L3U

WHEN?

9/14/2026 6:15:10 PM

WHAT EVENT?

4720 — User account created

That's a basic SOC investigation.

3. SOC investigation report:

Event ID: 4720

Time: 14 Sep 2026 18:15:10

Computer: DESKTOP-H4H3L3U

Created by: Shadow

New account: RootExploit

TaskCategory = User Account Management

Keywords = Audit Success

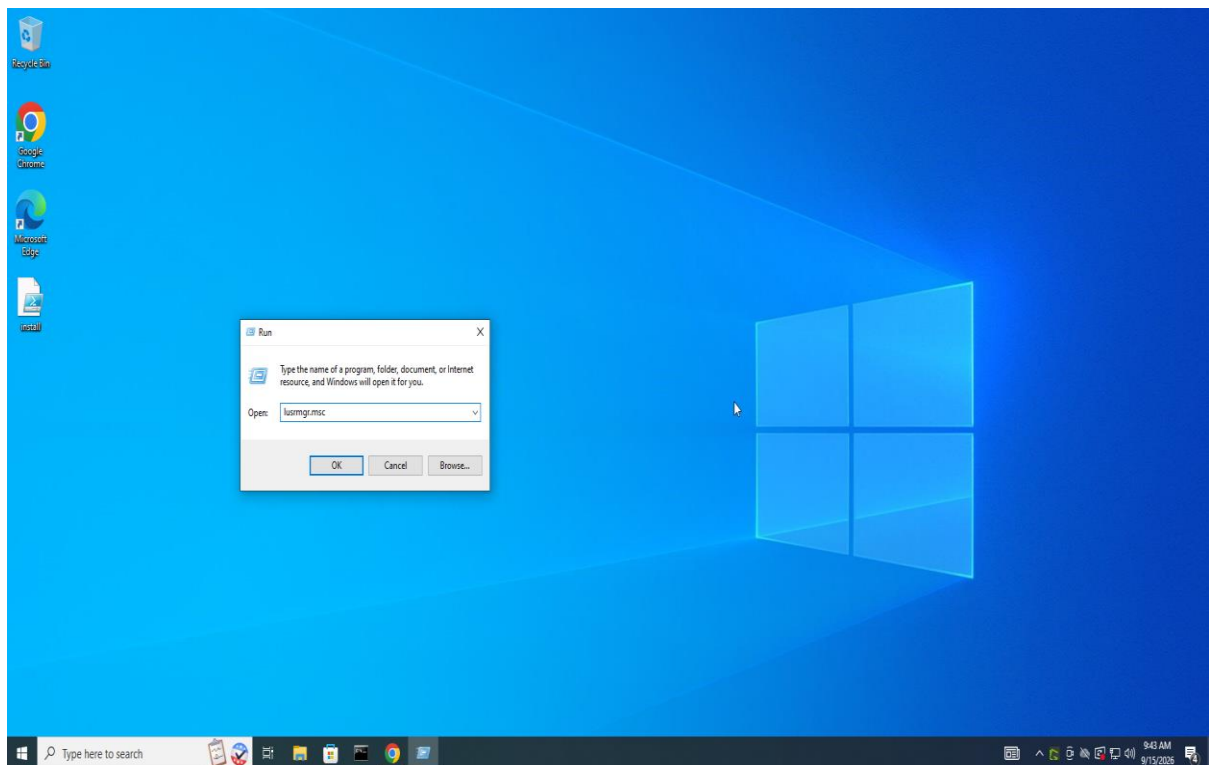
Phase-7: Detection-3. Privilege/group modification

Step-38: We'll add your test user RootExploit to the local Administrators group. Windows should generate Event ID 4732.

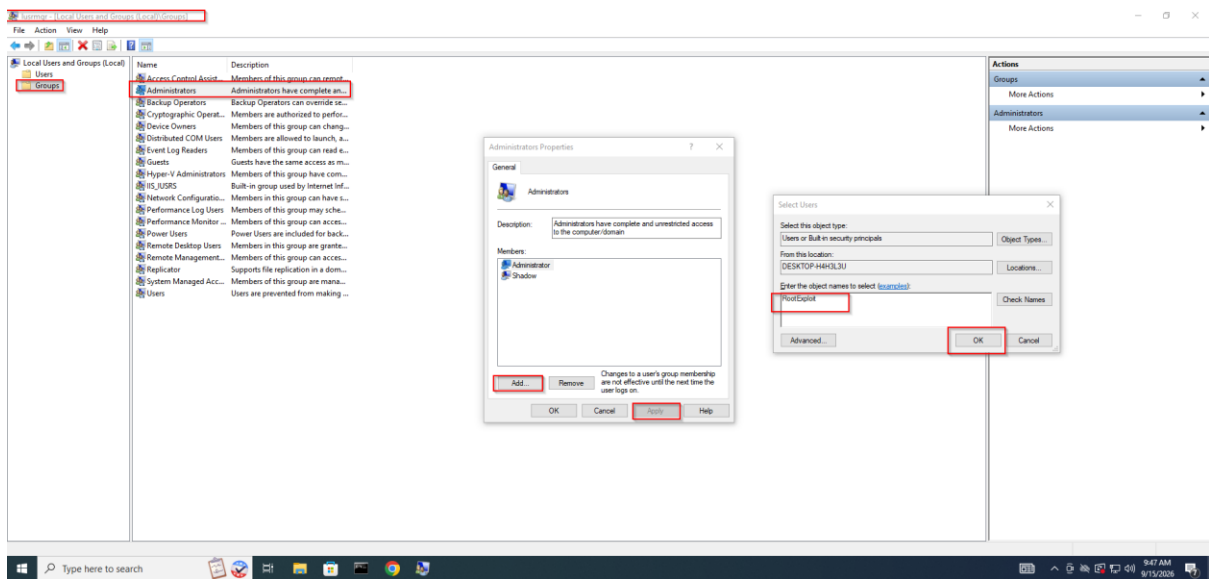
1. Add RootExploit to Administrators

On your Windows VM:

1. Press **Win + R**
2. Type: `lusrmgr.msc`



3. Select **Groups**
4. Double-click **Administrators**
5. Click **Add**
6. Enter: RootExploit
7. Click **Check Names**
8. Click **OK**

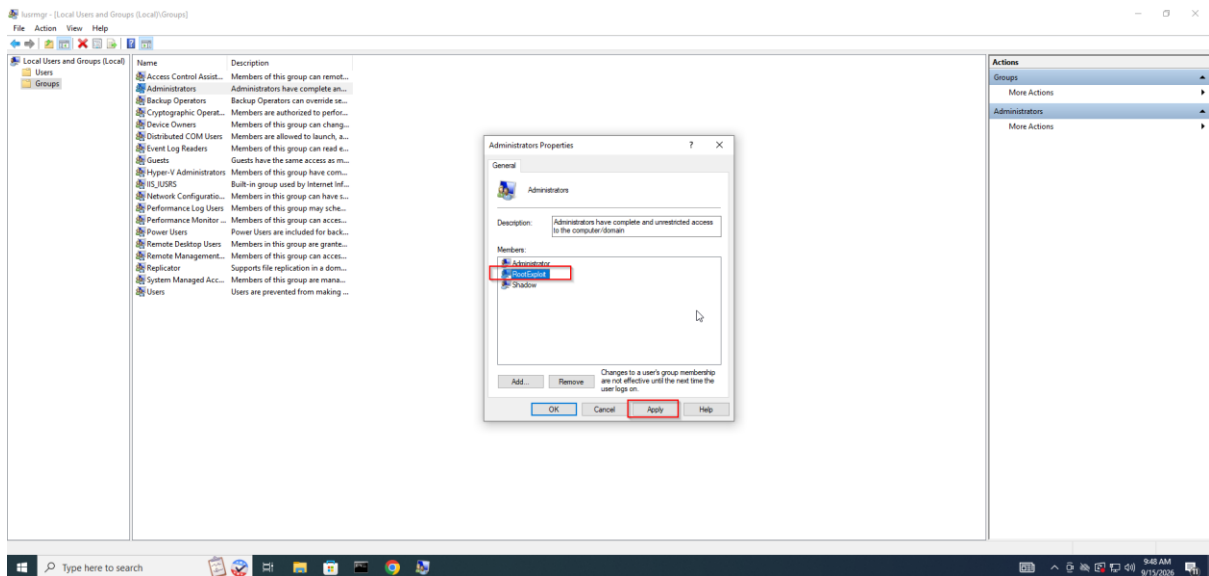


7. Apply → OK

Now:

Administrators

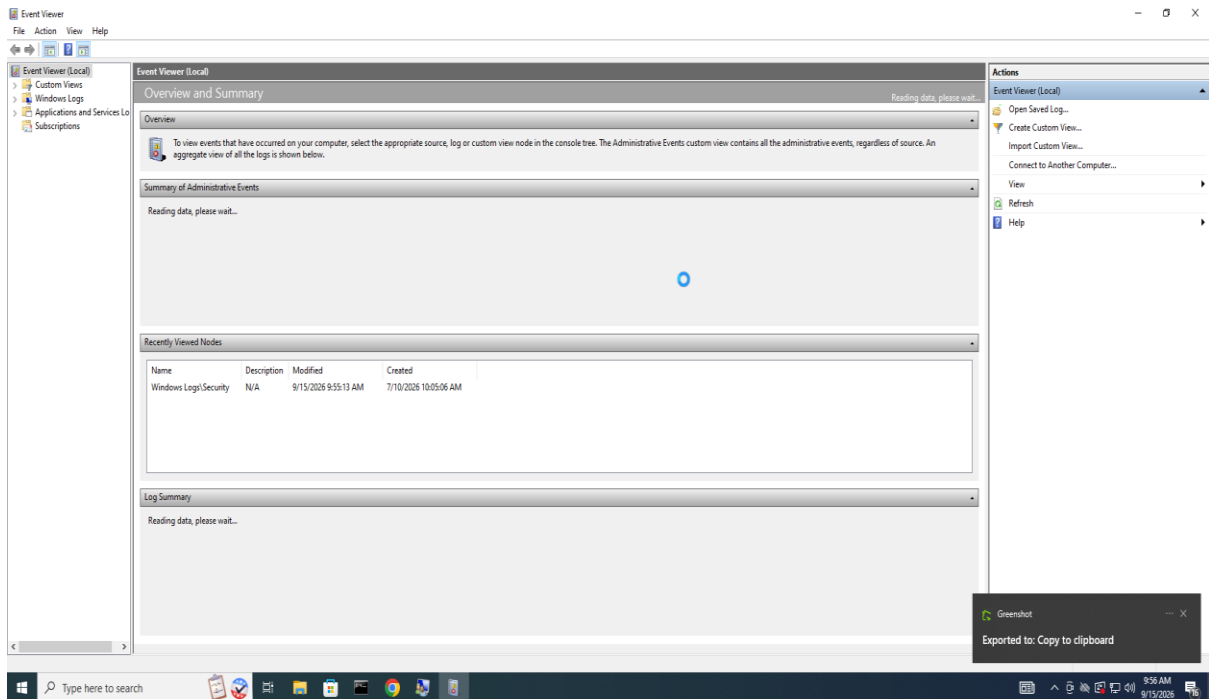
|
└─ RootExploit



You've just created the privilege/group modification for your lab.

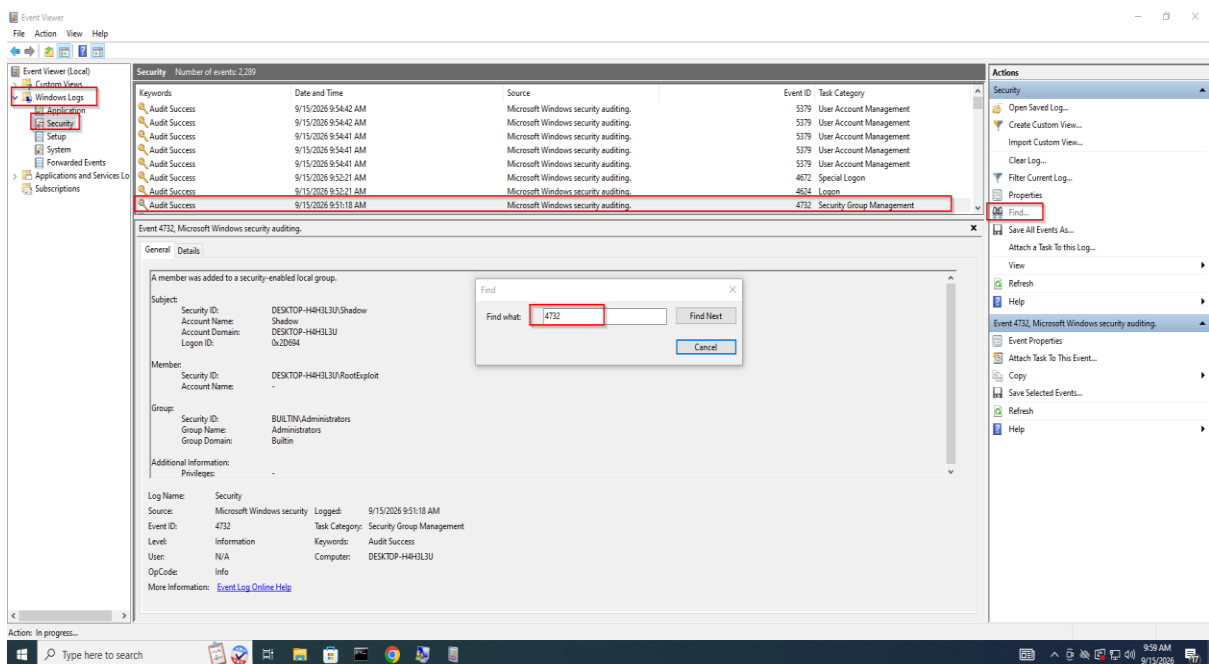
2. Check Windows Event Viewer

1. Open: eventvwr.msc



2. Go to: Windows Logs → Security

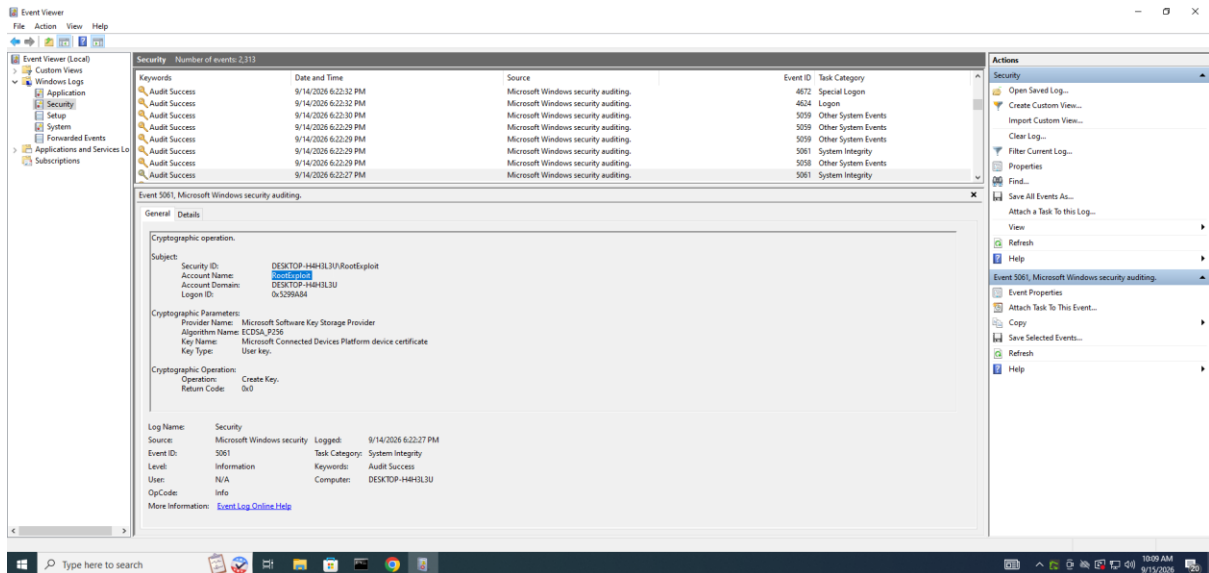
Look for: Event ID: 4732



3. A member was added to a security-enabled local group.

Group Name: Administrators

Member Name: RootExploit



3. Find it in Splunk

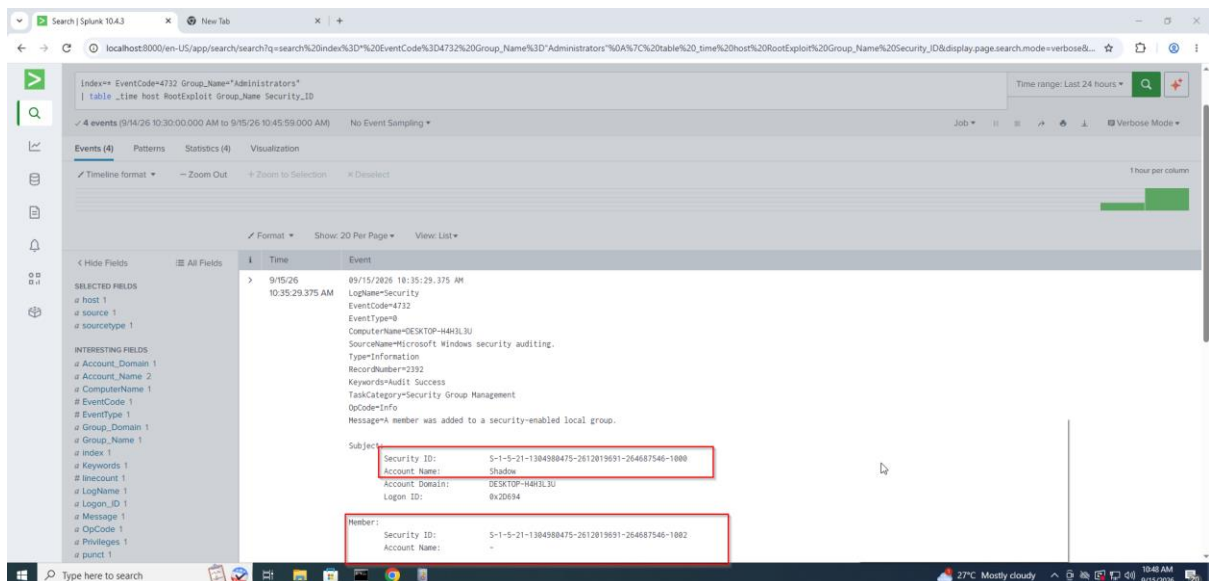
1. Click the Search bar

Command: index=* EventCode=4732 Group_Name="Administrators"

| table _time host Subject_Account_Name Member_Security_ID Group_Name

Working: Splunk search found 4 Event ID 4732 events. The highlighted event confirms:

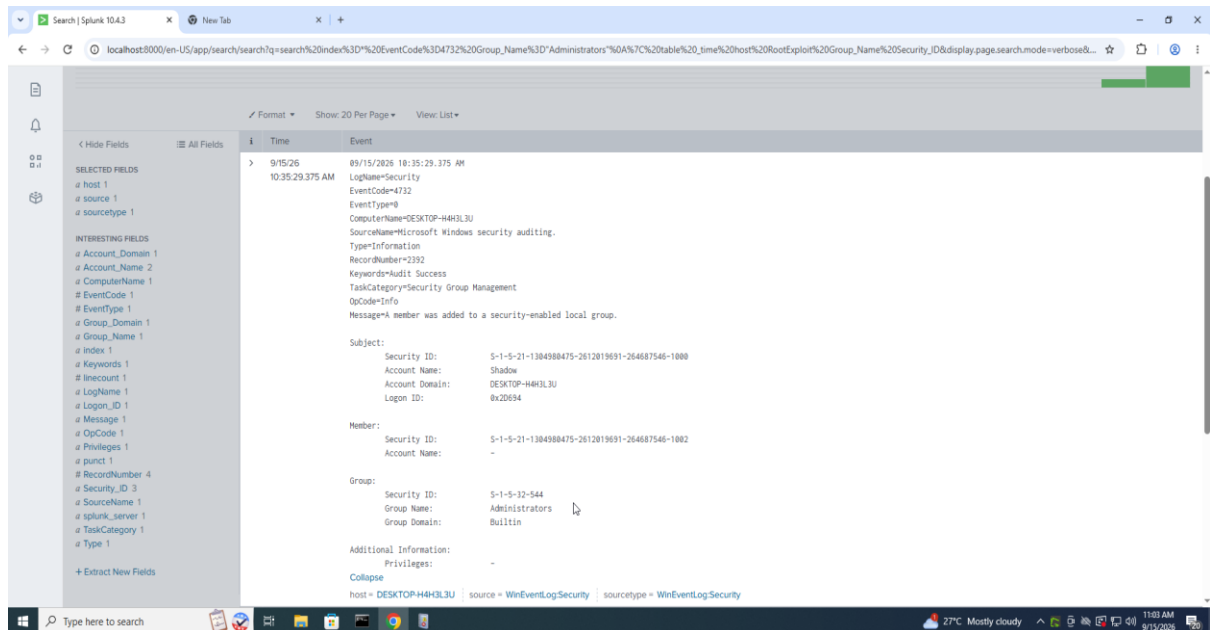
Shadow → added RootExploit (...-1002) → Administrators group



2. EventCode = 4732

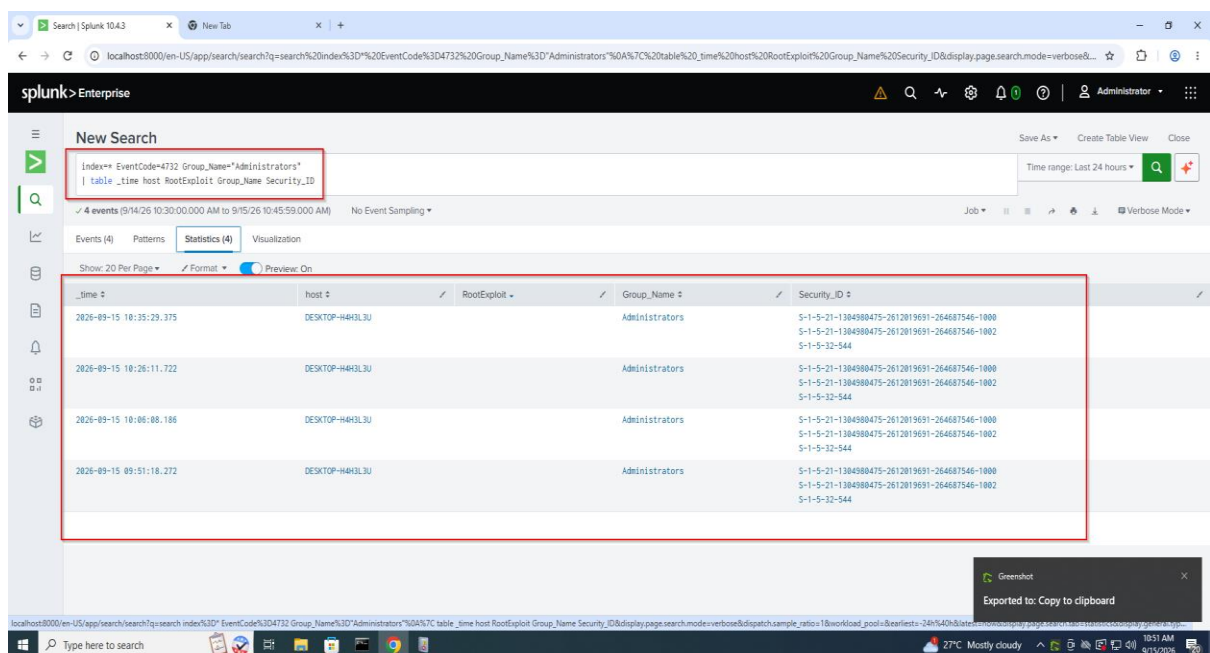
- Account Name = Shadow
- Member SID = ...-1002 (RootExploit)
- Group Name = Administrators

Detection is successfully working.



3. Search returned 4 Event ID 4732 events for the Administrators group.

Detection confirmed: Shadow → RootExploit → Administrators



For this event, the SOC analyst checks 5 things:

1. **Who performed the action?** → Shadow
2. **Who was added?** → RootExploit (SID ...-1002)
3. **Which group?** → Administrators
4. **When/where?** → timestamp + DESKTOP-H4H3L3U
5. **Was it authorized or suspicious?** → investigate why Shadow added RootExploit.

Investigation should become:

Shadow

| **added**



RootExploit



Administrators



Event ID 4732

This is important because an attacker who gains access to a normal account may try to add another account to **Administrators** to obtain elevated privileges.

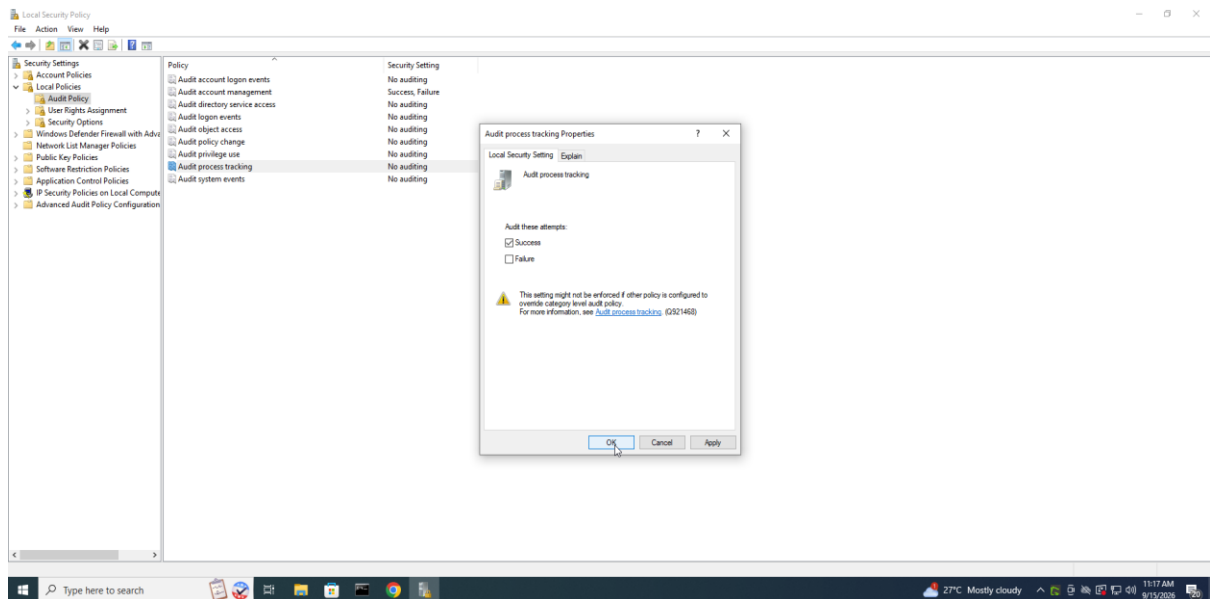
Phase-8: Detection-4 – PowerShell

Step-39: Event ID 4688 — Process Creation

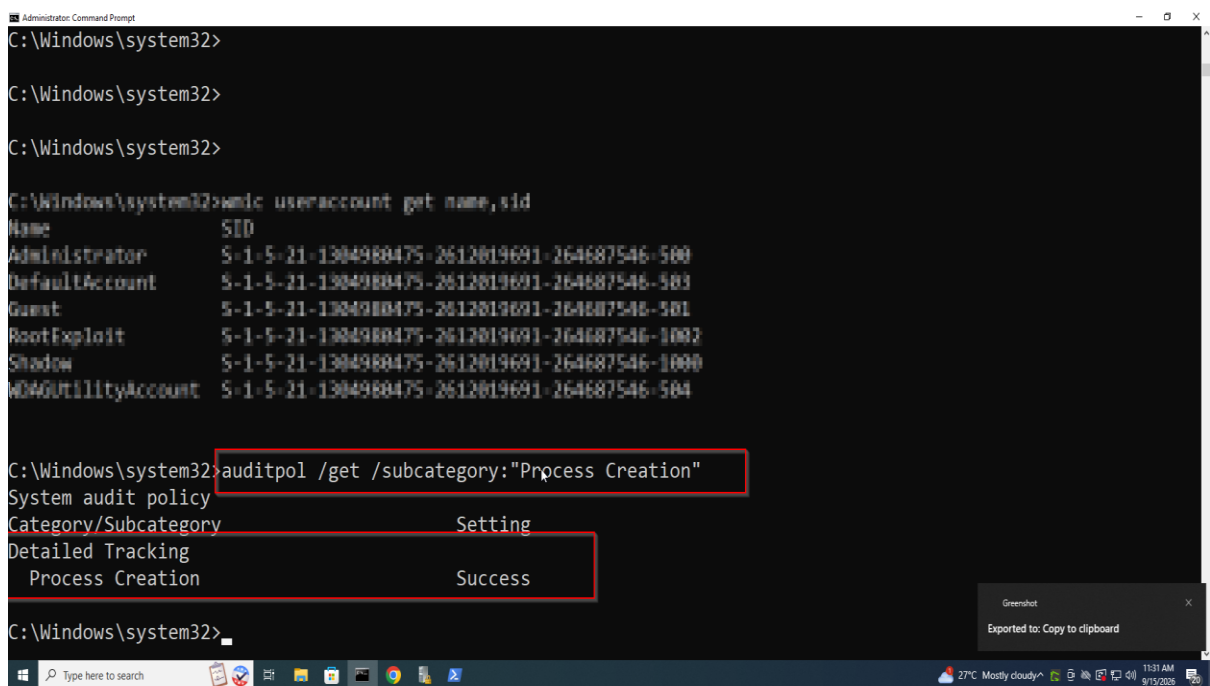
1. Enable process creation auditing

Advanced Audit Policy Configuration-System Audit Policies-Detailed Tracking-Enable: Success.

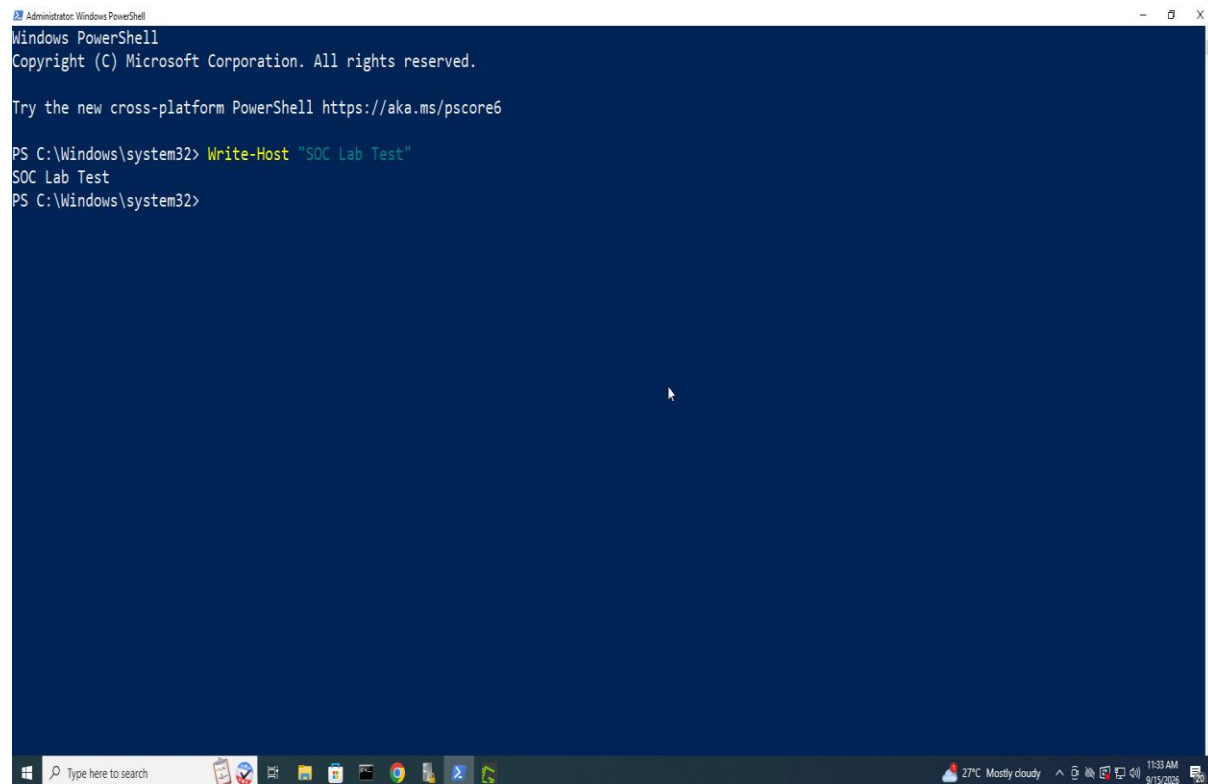
Click **Apply** → **OK**.



2. Enable command-line auditing



3. Generate a PowerShell event

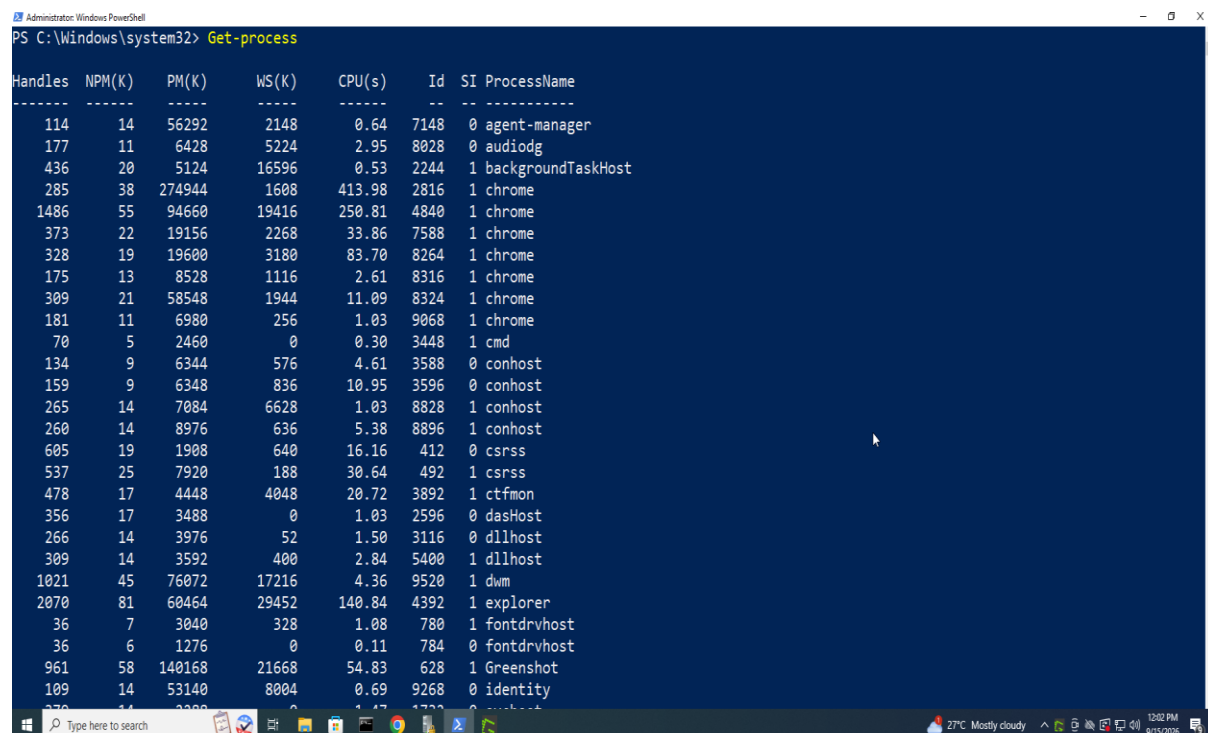


```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> Write-Host "SOC Lab Test"
SOC Lab Test
PS C:\Windows\system32>
```

4. This should generate process creation events.



```
Administrator: Windows PowerShell
PS C:\Windows\system32> Get-process
```

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
114	14	56292	2148	0.64	7148	0	agent-manager
177	11	6428	5224	2.95	8028	0	audiodg
436	20	5124	16596	0.53	2244	1	backgroundTaskHost
285	38	274944	1608	413.98	2816	1	chrome
1486	55	94660	19416	250.81	4840	1	chrome
373	22	19156	2268	33.86	7588	1	chrome
328	19	19600	3180	83.70	8264	1	chrome
175	13	8528	1116	2.61	8316	1	chrome
309	21	58548	1944	11.09	8324	1	chrome
181	11	6980	256	1.03	9068	1	chrome
70	5	2460	0	0.30	3448	1	cmd
134	9	6344	576	4.61	3588	0	conhost
159	9	6348	836	10.95	3596	0	conhost
265	14	7084	6628	1.03	8828	1	conhost
260	14	8976	636	5.38	8896	1	conhost
605	19	1908	640	16.16	412	0	csrss
537	25	7920	188	30.64	492	1	csrss
478	17	4448	4048	20.72	3892	1	ctfmon
356	17	3488	0	1.03	2596	0	dasHost
266	14	3976	52	1.50	3116	0	dllhost
309	14	3592	400	2.84	5400	1	dllhost
1021	45	76072	17216	4.36	9520	1	dwm
2070	81	60464	29452	140.84	4392	1	explorer
36	7	3040	328	1.08	780	1	fontdrvhost
36	6	1276	0	0.11	784	0	fontdrvhost
961	58	140168	21668	54.83	628	1	Greenshot
109	14	53140	8004	0.69	9268	0	identity
270	14	3080	0	1.47	1720	0	notepad

5. Search Event ID 4688 in Splunk

The screenshot shows the Splunk Enterprise interface with a search for Event ID 4688. The search bar contains the query `index=* EventCode=4688`. The results show 436 events from 9/14/26 12:30:00.000 PM to 9/15/26 1:05:35.000 PM. The event details for EventCode=4688 are displayed, including fields like `LogName=Security`, `EventCode=4688`, `EventType=8`, `ComputerName=DESKTOP-H4H3L3U`, `SourceName=Microsoft Windows security auditing`, `Type=Information`, `RecordNumber=32147`, `KeyWord=audit: Success`, `TaskCategory=Process Creation`, `OpCode=Info`, and `Message=A new process has been created.`

The screenshot shows the Splunk Enterprise interface with a search for Event ID 4688. The search bar contains the query `index=* EventCode=4688`. The results show 436 events from 9/14/26 12:30:00.000 PM to 9/15/26 1:05:35.000 PM. The event details for EventCode=4688 are displayed, including fields like `LogName=Security`, `EventCode=4688`, `EventType=8`, `ComputerName=DESKTOP-H4H3L3U`, `SourceName=Microsoft Windows security auditing`, `Type=Information`, `RecordNumber=32147`, `KeyWord=audit: Success`, `TaskCategory=Process Creation`, `OpCode=Info`, and `Message=A new process has been created.`

What a SOC analyst checks for PowerShell:

A SOC analyst asks:

Who?

Account Name

What?

powershell.exe

When?

_time

Where?

host

What command?

CommandLine

Which parent process?

ParentProcessName

Why was it executed?

This is the most important question.

For example:

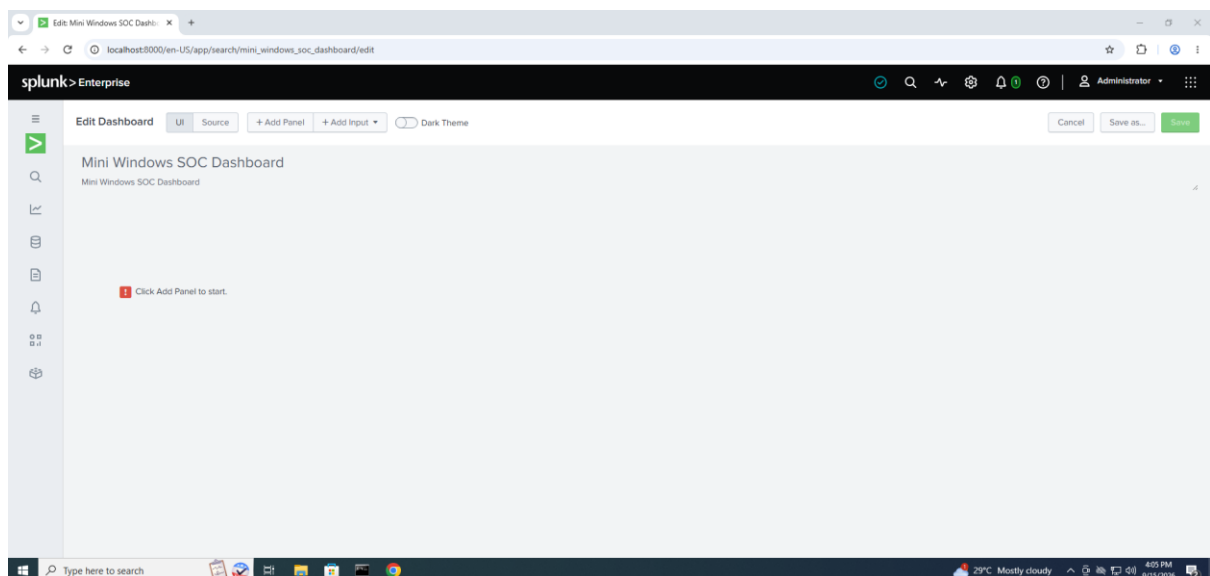
powershell.exe by itself may be completely normal deserves investigation.

Phase-9: Build a Splunk dashboard

After the detections work, we'll make a small SOC dashboard.

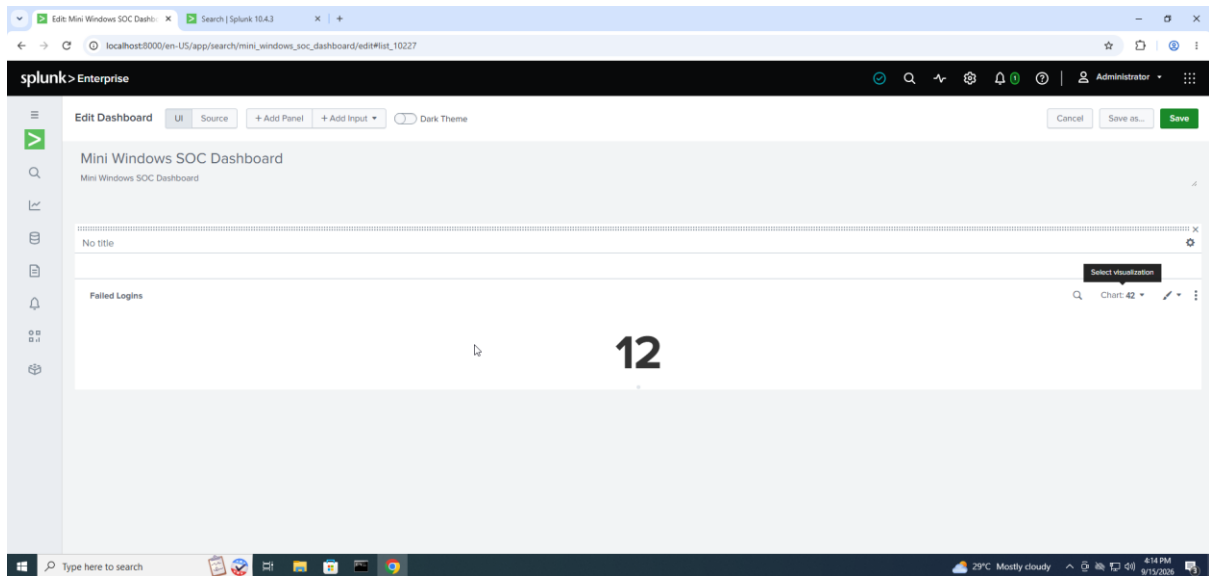
1. Open Dashboard creation

In Splunk: **Dashboards** → **Create New Dashboard**



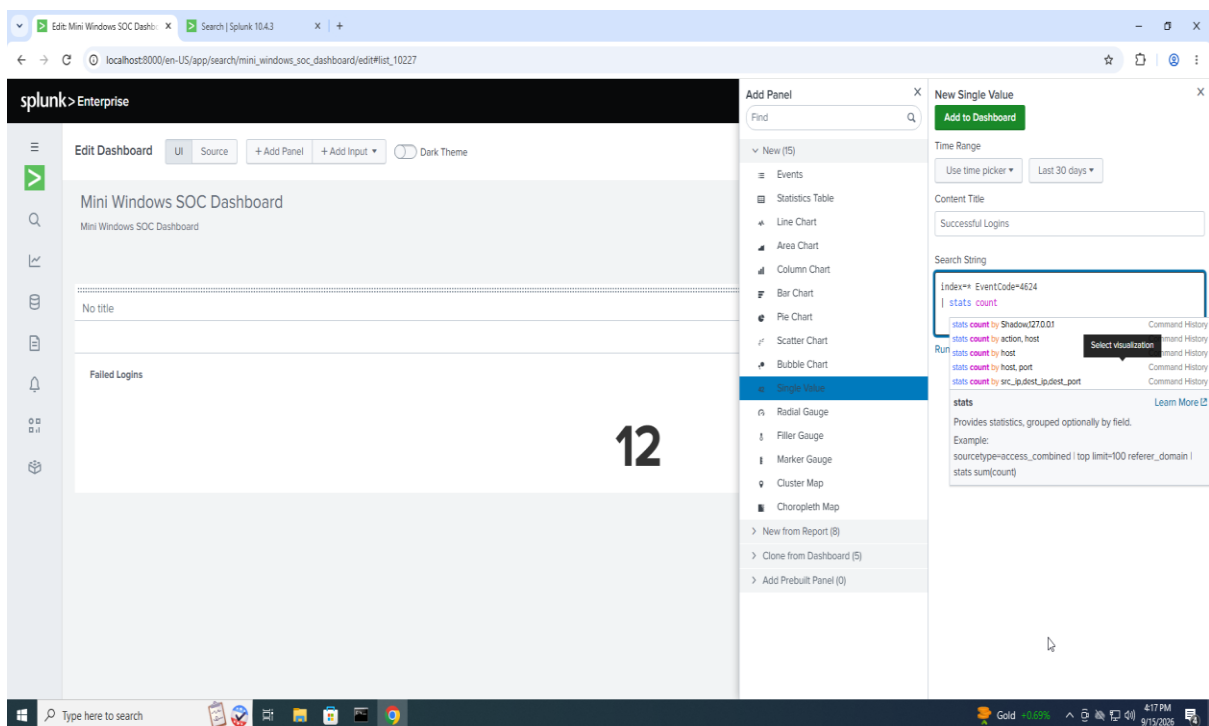
2. Panel 1 - Failed Logins

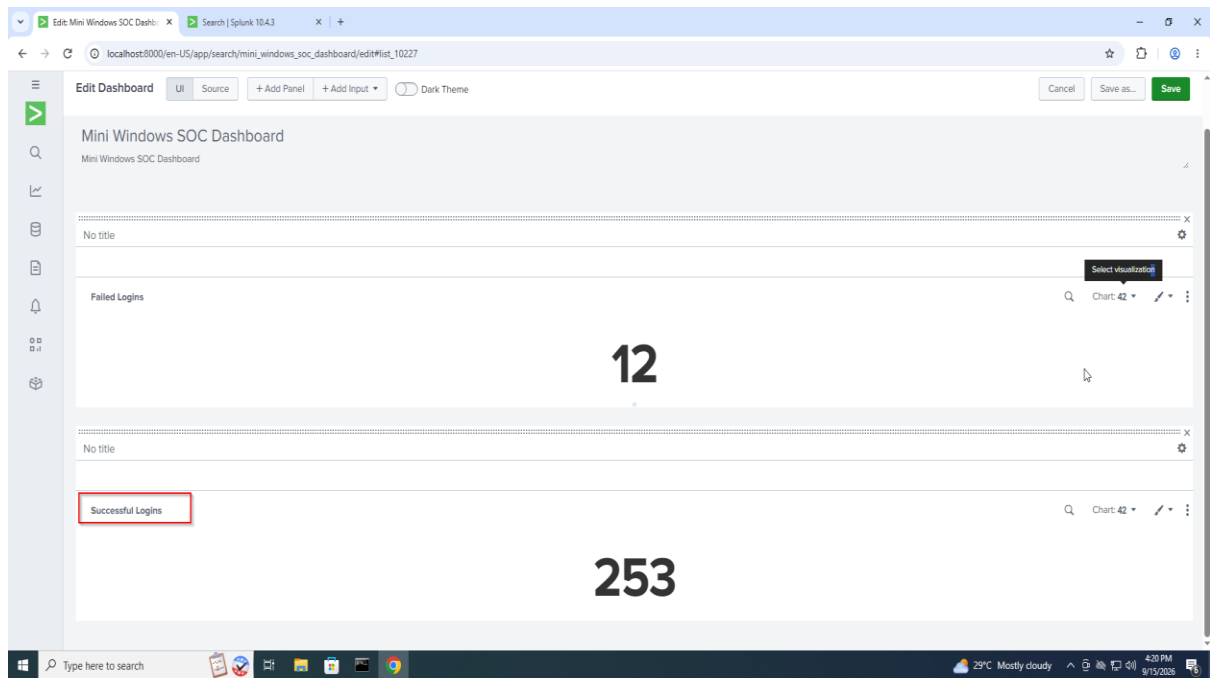
This is your 4625 detections. Create a new panel and use:



3. Panel 2 - Successful Logins

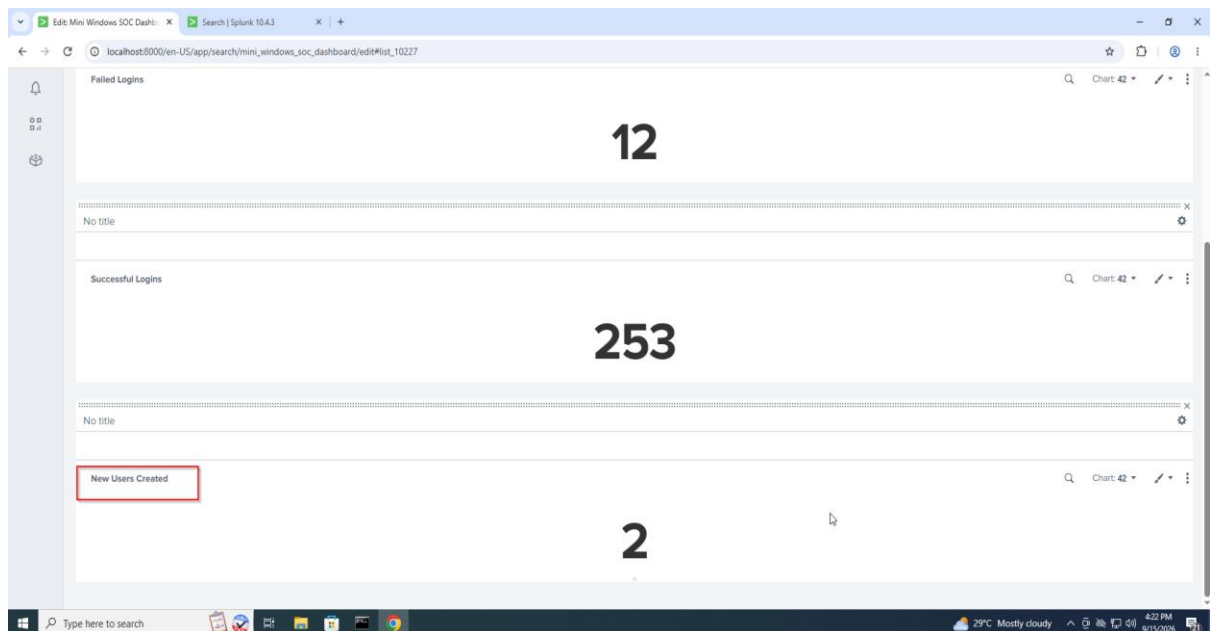
Use Event ID 4624.





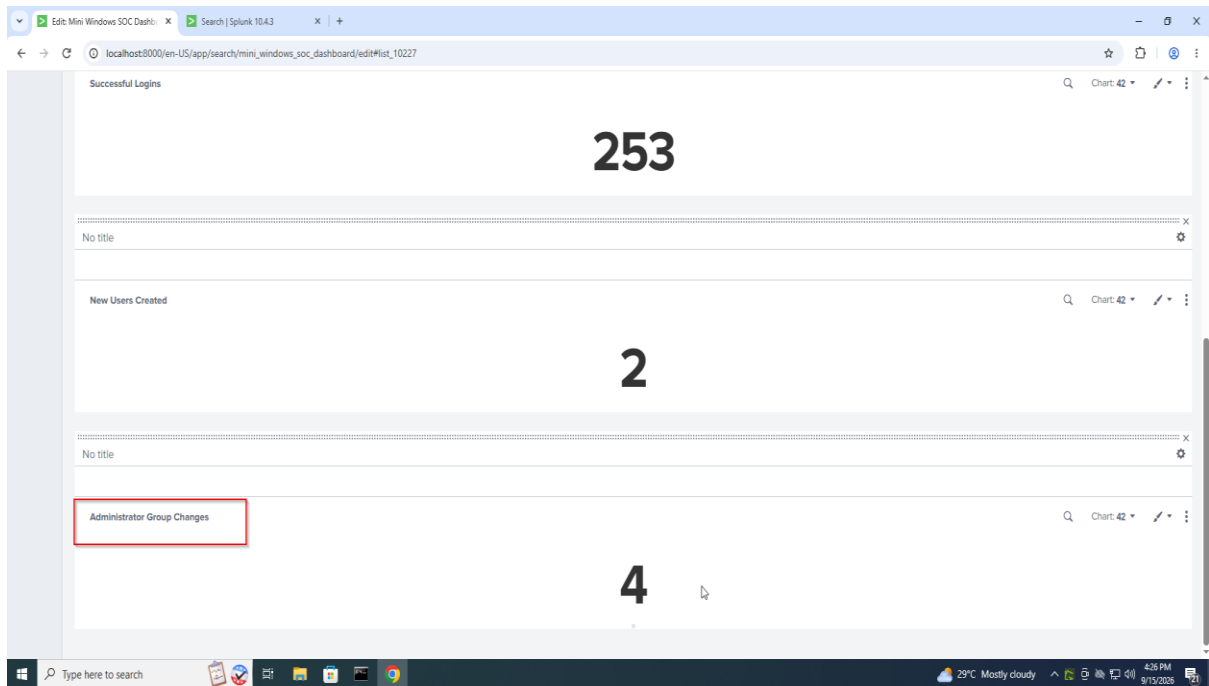
4. Panel 3 - New Users

We are tested Event ID 4720.

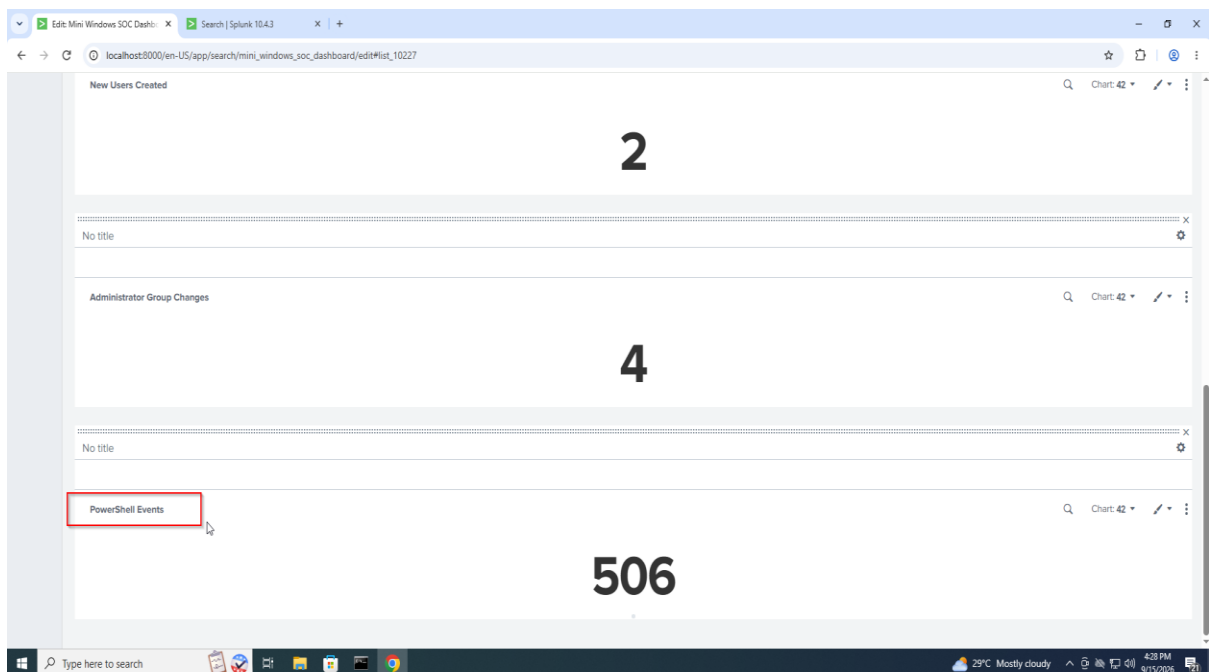


5. Panel 4 - Administrator Group Changes

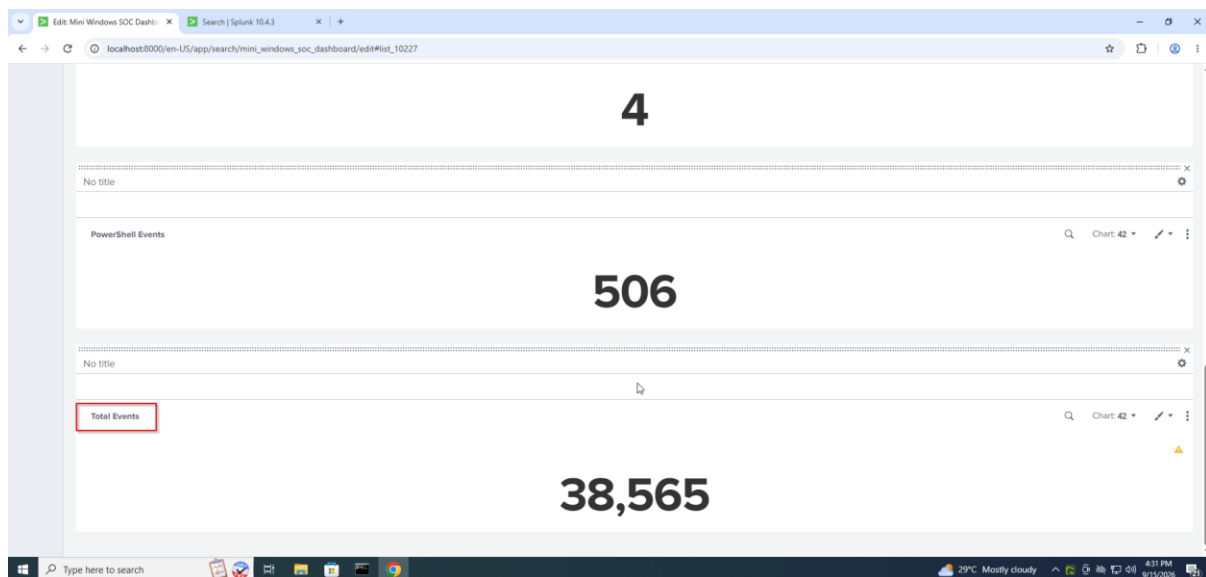
This is an important SOC metric because it shows accounts being added to the local Administrators group.



6. Panel 5 - PowerShell Events



7. Panel 6 - Total Windows Events



Conclusion:

This Mini SOC project provided practical experience in Windows log monitoring, event analysis, threat detection, and security investigation using Splunk. The project demonstrated how Windows security events can be collected, monitored, and analyzed to identify potentially suspicious activities.

During the project, important Windows Event IDs such as **4625 (failed logon)**, **4720 (new user account)**, **4732 (group membership modification)**, and **4688 (process creation)** were investigated. Splunk was used to search and analyze these events and create detection queries and dashboard panels.

The project also demonstrated a basic SOC investigation approach by identifying **who performed an action, what happened, when and where it occurred, and why the activity may be suspicious**. This helped connect raw Windows events with practical security investigations.

Overall, the project strengthened practical skills in **Windows Event Monitoring, Splunk SIEM, SPL queries, threat detection, log analysis, incident investigation, and SOC operations**. It provides a strong foundation for further improvements such as advanced detection rules, MITRE ATT&CK mapping, alert automation, and real-world incident response workflows.